J.H. CERILLES STATE COLLEGE
School of Computing Studies
Bachelor of Science in Information Technology
Mati, San Miguel, Zamboanga del Sur
Main Campus

Darb e-Link System

A CAPSTONE PROJECT

Presented to the faculty of

School of Engineering and Technology

In Partial Fulfilment

Of the Requirements for the Degree

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

By:

Roda, Ian Christopher T.

Labad, Junelie F.

Cabog, Mark Cloyd A.

Porferio, Melvin A.

2026

APPROVAL SHEET

The Capstone Project entitled **Darb e-Link System** prepared and submitted by, Ian Christopher T. Roda, Junelie F. Labad, Mark Cloyd A. Cabog, and Melvin A. Porferio, has been examined and is recommend for approval and acceptance.

RECOMMENDED:

MARVIN C. MEJORADA
Adviser

APPROVED by the IT Project Defense Panel with a grade of _____.

EDILBERTA G. NAPARAN, MSIT
Chairperson

MARY DAME ANGELICA M. CANILLO, MSIT
Panel Member

DAYAHAN A. BUGAO, MSIT
Panel Member

ACCEPTED and APPROVED in partial fulfillment of the requirements in Bachelor of Science and Information Technology.

MARIBEL M. ESTIOCO, MST-CS
BSIT Program Chair, Main

Campus Date: _____

ACKNOWLEDGEMENT

The researchers wish to express their heartfelt gratitude to the following individuals who played essential roles in completing this study.

To their adviser Mr. Marvin C. Mejorada and IT capstone instructor, Mrs. Maribel M. Estioco, for her unwavering patience, innovative ideas, expert guidance, continuous support, extensive assistance, and motivating encouragement throughout the entire research journey.

To their panelists, Mrs. Edilberta G. Naparan, Mrs. Mary Dame Angelica M. Canillo, and Mr. Dayahan Bugao, for their valuable insights, constructive feedback, thorough corrections, professional opinions, and practical suggestions that strengthened and refined this study.

To their parents, guardians, and loved ones, for their steadfast love, financial backing, moral encouragement, understanding patience, and inspiring motivation that sustained them through challenges and helped realize their aspirations.

Finally, to Almighty God, for the divine wisdom, inner strength, and abundant blessings that empowered them to persevere and successfully complete this work.

The researchers are profoundly thankful for each person's invaluable contributions to their achievement.

Researchers

EXECUTIVE SUMMARY

The study entitled “Darb e-Link System: A Web-Based Online Food Ordering Platform for Small Food Vendors at J.H. Cerilles State College Main Campus” was conducted to solve the problem of slow and unorganized food ordering among small vendors on campus. Currently, vendors use manual methods such as social media messages and handwritten lists to take orders, which cause delays, wrong orders, confusion, and long waiting times, especially during busy hours. The main objective of this project was to develop a web-based system that makes ordering easier and faster for students, faculty, and staff. The system allowed users to view menus, place orders for pickup or delivery, choose payment methods like GCash, Maya, or Cash on Delivery, and receive SMS updates about their order status. The study used the 4D Model (Discover, Design, Develop, Deliver) in developing the system, which included data gathering, system design, development, testing, and implementation. The system was built using web technologies to make it accessible on phones and computers. The developed system provided a simple and organized way of handling orders, which helped reduce errors, save time, and improve the overall experience of both customers and vendors.

TABLE OF CONTENTS

Content	Page
APPROVAL SHEET	ii
ACKNOWLEDGEMENT	iii
EXECUTIVE SUMMARY	iv
TABLE OF CONTENTS	v
CHAPTER I: BACKGROUND OF THE STUDY	
Introduction	1
Project Context	2
Purpose and Description	3
Objectives of the study	4
Scope and Limitations	4
Definition of Terms	5
CHAPTER II: REVIEW OF RELATED STUDIES AND SYSTEMS	
Related Studies and Systems	9
Foreign	9
Local	10
CHAPTER III: DESIGN AND METHODOLOGY	
Design and Methodology	14

Requirements Analysis 14

Discover Phase 14

Design Phase 15

Develop Phase 20

Deploy Phase 21

CHAPTER IV: DEVELOPMENT AND TESTING

Description and Prototype 22

Development 23

Menu Management Module 23

Noticafation Module 25

Testing

Alpha Testing 27

Compatibility Checking 28

CHAPTER V IMPLEMENTATION RESULT, DISCUSSION, CONCLUSION AND RECOMMENDATION

Implementation Result and Discussion 31

Conclusion 46

Introductory Sentence 47

REFERENCES

APPENDICES

A. Test Cases 50

B. Sample Input and Output Report 58

C. Documentation 60

D. Survey Questionnaire 61

E. Curriculum Vitae 63

LIST OF FIGURES

Figures **Page**

1. 4D Model 13

2. Current Technical Situation 14

3. Context Diagram 15

4. Conceptual Framework Diagram 15

5. Activity Diagram 16

    a. Admin 16

    b. Customer 16

6. Use Case Diagram 17

7. Entity Relationship Diagram 19

8. System Structured Diagram 20

9. System Architecture Diagram 21

LIST OF TABLES

Tables **Page**

1. Hardware Specification 8
2. Software Specification 8
3. Use Case Description (Entry Food) 18
4. Use Case Description (Create Order) 18
5. Gantt Chart 22
6. Implementation Plan 28

Survey Questionnaire of the Darb e-Link System

7. All 41
8. Student 42
9. Staff 43
10. Faculty 45

LIST OF IMAGES

Images **Page**

1. Interview abd System Presentation 14
2. Alpha Testing 27
3. Implementation Result and Discussion

Menu 31

Cart 32

Checkout 32

Confirm Order 33

Order History 33

About 34

Contact Us 34

Login 35

Dashboard 35

Products 36

Orders 37

Categories 37

Chapter I
THE PROBLEM AND ITS SCOPE

Introduction

The growth of digital technology has significantly reshaped consumer behavior and business operations, particularly in the food service industry. Online food ordering systems enable customers to browse menus, customize orders, and complete transactions quickly and conveniently through web or mobile applications (Sethu & Saini, 2016). These systems enhance efficiency by reducing errors in manual ordering, streamlining communication between customers and establishments, and offering flexible payment options (Ray et al., 2019). As more consumers prefer accessible and time-saving services, the adoption of online ordering platforms has become an essential strategy for food businesses to remain competitive and responsive to market demands.

Developing an online food ordering platform is important because it enhances both customer experience and business efficiency. By providing a fast, accurate, and convenient way to place orders, it minimizes errors commonly associated with manual processes and reduces waiting time for customers (Chakraborty & Sur, 2019). Furthermore, an online system broadens market reach by appealing to consumers who value accessibility and convenience, making it a vital innovation for sustaining competitiveness in the food service industry.

To address these needs, the proposed Darb e-Link online ordering platform is designed to provide a seamless and reliable system for both customers and food providers. It offers essential features such as digital menus, accurate order processing, and secure payment methods, ensuring a smooth and convenient user experience. At the same time,

it assists food businesses by improving order management. With its user-centered design and operational efficiency, Darb e-Link stands as a practical and innovative solution that not only streamlines food ordering but also strengthens the competitiveness of food establishments in the digital marketplace.

Project Context

In today's fast-paced digital world, small businesses often struggle to manage customer orders efficiently. Traditional ordering methods, like messaging apps, manual tallying, can cause delays, errors, and miscommunication between sellers and customers. These problems are worse during peak hours when handling multiple orders.

Inside the campus, small vendors like Darb still do not have access to big e-commerce platforms. Darb serves not only students but also teachers, staff, and visitors inside the school. At present, they mostly use social media to take orders, but this often leads to confusion, delays, and missed transactions. The Darb e-Link System provides a simple and reliable way to handle orders, helping Darb and other campus vendors run their business in a faster, more organized, and professional manner.

Every day, many students, employee and staffs buy from Onyx, especially during break times. However, since the ordering process is still manual, problems such as mix-ups, missed orders, and longer waiting times often occur when many customers order at once. The Darb e-Link System was developed to solve these challenges. It is designed mainly for Darb but can also be used by other small vendors who face the same struggles. This system also supports multiple payment options such as online payments through GCash and Maya, or cash-on-delivery (COD) for

customers who prefer to pay upon receiving their orders. For online payments, customers are required to upload their receipt as proof of payment. On the other hand, for COD transactions, the delivery personnel will take a photo as proof of payment received. In addition, the system offers a pickup feature that allows customers to place an order online and simply pick it up once it is ready. With these features, orders can be managed more smoothly, quickly, and professionally, giving both sellers and customers a better overall experience.

Purpose and Description

The goal of this project is to create an easy-to-use, link-based online food ordering system for customers. Through a unique link, customers can access an order form, browse available food items, and place their orders using a responsive and mobile-friendly interface. This makes the ordering process faster and reduces the need for face-to-face or manual communication.

The system includes product listings with images and prices, a dynamic order form, order status updates, and an admin dashboard to manage the menu and orders. Customers are notified through SMS when their order is confirmed, out for delivery, ready for pickup, or delivered. For online payments through GCash and Maya, customers are required to upload their payment receipt as proof, which the admin reviews before completing the order. For cash-on-delivery (COD), the delivery personnel will take a photo as proof of payment received to ensure proper documentation. This provides customers with more flexible payment options while maintaining secure and reliable transactions.

Overall, the Darb e-Link System provides a smooth and organized ordering process for customers while giving the admin an easy way to manage menus and track orders.

Customers can choose delivery or pickup, making transactions more convenient, while the admin ensures that orders are processed correctly. This project aims to improve food ordering for Darb by making it more efficient and digital.

Objectives of the study

General Objectives

The proponents aim to develop a web-based link order system to enable customers to conveniently place orders online and assist sellers in managing orders efficiently.

Specific Objectives

This proposed system aims to:

1. Design a web-based link order system that lets customers place orders using a digital form with product options.
2. Develop a simple and efficient backend dashboard for sellers to monitor and manage incoming orders in real time.
3. Test the functionalities, the system regularly assist the operation of the environment.

Scope and Limitations

The Darb e-Link System is designed for small to medium-sized businesses that need a simple digital ordering solution. The admin can upload products, set prices, and create unique order links to share online. Customers can browse products, place orders, and get status updates without needing to create an account. The system includes core functions like product listing, order submission, and status tracking, and works on both desktop and mobile browsers. The admin manages orders and updates statuses through a web-based panel.

For online payments, the system is not directly connected to the GCash or Maya applications. Customers are required to manually upload a screenshot or receipt of their transaction as proof of payment. The admin verifies the uploaded proof before processing the order. For cash-on-delivery (COD), the delivery personnel will take a photo as proof of payment received to ensure proper documentation.

The system has some limitations. It is built for single-store use and does not support multiple vendors. Additionally, since receipt uploads depend on customers submitting correct and clear proof of payment, delays may occur if the uploaded receipt is missing or incorrect.

Definition of Terms

This part explains the specific meaning of important words as they are used in this study.

Darb e-Link System: This term refers to the title of the proposed web-based platform designed to create and manage online forms through unique product links. It enables customers to place orders digitally while allowing sellers to efficiently monitor and process these orders in real time.

Link: This term refers to a unique URL that directs customers to the digital menu, where they can browse available items and place their orders online. It serves as the access point for customers to view the full menu and make their selections.

Dashboard: This term refers to the web interface used by sellers to manage orders. It allows them to update order statuses, monitor customer activity, and efficiently handle incoming orders through a centralized control panel.

Responsive Design: This term refers to the web design approach implemented in the system to ensure optimal functionality and user experience across various devices such as mobile

phones, tablets, and desktop computers.

Customer Order: This term refers to a submission made by a customer through the Link, indicating their selected products and quantities. It represents the formal request processed by the system for fulfillment.

Users: This term refers to the individuals who interact with the Darb e-Link System. It includes both customers, who place orders through the Link, and sellers, who manage and process these orders via the Dashboard.

CHAPTER II
REVIEW OF RELATED STUDIES AND SYSTEMS

This chapter presents the technical background of the project, including the hardware and software used, the programming tools, and the related foreign and local studies that support the idea behind the Darb e-Link System.

Technical Background

The Darb e-Link System is an online ordering platform that enables customers to place orders via a simple, unique link. This system is particularly beneficial for small businesses, such as food stalls and online sellers, as it allows them to manage orders efficiently without relying on complicated applications. By streamlining the ordering process, it minimizes the chances of human error and helps prevent lost orders (Egereonu, 2024).

Online ordering systems have become widely adopted by both small and large businesses. They assist sellers in showcasing products online, receiving customer orders, and providing timely updates. Common features include product listings, order forms, and administrative dashboards. Thanks to simple web technologies, sellers can manage orders using their phones or basic computers, making these systems accessible and convenient (Peng et al. 2024).

The platform also utilizes a database to securely store orders and product details. Databases consolidate information in one location, making it easy to access and update. The combination of a user-friendly web interface and a reliable database ensures that the system is efficient and accessible for both sellers and customers (Acharya 2024 & Alcalde et al. 2022).

Hardware Specifications

Hardware	**Description**
Desktop/Computer	Intel Core i3
Hard Disk Drive	300GB
Processor	Core i3
RAM	4GB

Software Specifications

Software	**Description**
Server Environment	Windows 10
Browser	Google Chrome
XAMPP	v3.3.0
Code Editor	Visual Studio Code v1.84
Client Browser	Google Chrome
Hosting (Optional)	InfinityFree (1GB storage)

Programming Environment

The system creates an interactive and responsive web interface using HTML, CSS, and JavaScript. While MySQL acts as the database to safely store product, order, and customer information, PHP is used on the backend to manage data processing and server-side operations. These technologies were selected for the purpose of creating an effective and user-friendly web-based ordering system because of their dependability, flexibility, ease of use, and strong integration.

Related Studies

Foreign

Optimized Web-Based Online Food Ordering System

From the International Journal of Computer Science Languages, Egereonu (2024) developed an online ordering system using a WAMP (Windows, Apache, MySQL, PHP) setup. The system targets small food businesses by enabling real-time customer order acceptance, delivery management, and report generation. The study found that the system improved order accuracy, streamlined daily operations, and increased customer satisfaction and loyalty by simplifying order processing and reducing errors.

This study is strong in real-time order acceptance, delivery management, and report generation, which improve order accuracy and operations. However, it is more suited for general food businesses and lacks simple, mobile-friendly features. The Darb e-Link System fills this gap by offering easy link-sharing, SMS updates, and a lightweight dashboard for campus vendors.

Online Food Ordering and Delivery: A Study on the Use of Customer Service Data and Quality Function Deployment

Peng, Erkoc, and Omachonu (2024) conducted a peer-reviewed study published in the Journal of Food Distribution Research that analyzes how customer service feedback can be integrated into online food delivery systems using the Quality Function Deployment (QFD) model. Their research focuses on leveraging real-time customer preferences to enhance service attributes such as timeliness, accuracy, and user interface design. The findings emphasize that data-driven improvements lead to higher customer satisfaction, reliability, and operational efficiency in modern food ordering platforms.

Its strength is using customer feedback to improve timeliness, accuracy, and design, leading to higher satisfaction. The gap is that it is complex and better for large platforms. The Darb e-Link System, in contrast, is simple and affordable, designed for small campus-based sellers.

Online Food Order Systems

Acharya (2024) presented a web-based food ordering platform available on SSRN designed to simplify the ordering process for customers and restaurant staff. The system features a dynamic user interface where customers can browse menus, select meals, and complete transactions online. It also provides restaurant personnel with tools to manage orders, update menus, and monitor sales. The study highlights how digital food ordering systems improve operational efficiency, boost customer satisfaction, and support evolving business needs through automation and real-time interaction.

The system provides complete ordering flow and management tools, which help large restaurants. The gap is that it does not focus on small vendors. The Darb e-Link System addresses this by adding SMS notifications, flexible payments (GCash, Maya, COD), and a pickup option fit for small sellers in schools.

Local

Customer Satisfaction of DLSUD Students on Online Food Service Delivery

Alcalde et al. (2022) from De La Salle University – Dasmariñas assessed how students perceived the service quality of online food delivery platforms during the pandemic. Using quantitative methods and the SERVQUAL model, the study measured customer satisfaction across five dimensions: reliability, assurance, tangibles, empathy, and responsiveness. Results showed high satisfaction with packaging quality, communication, and order accuracy, while delivery wait times and

personalized attention needed improvement. This study highlights the increasing reliance on food delivery apps and recommends service enhancements to better serve the youth market in the Philippines.

This study is strong in showing high satisfaction in packaging, communication, and order accuracy. The gap is that issues like long delivery wait times and lack of personalized service remained. The Darb e-Link System helps reduce waiting time through SMS updates and a pickup option, making the process faster for campus buyers.

Online Food Delivery Service Satisfaction, Food Choices, and Usage Frequency Among UPLB Students

Arias and Bayna-Mariano (2024) from the University of the Philippines Los Baños examined how students use and rate online food delivery services. The study found that convenience, food appeal, and service quality significantly influence satisfaction and frequency of use. It emphasizes the need to improve online food platforms to meet the evolving preferences of young consumers, especially during the pandemic. The study highlights the strength of convenience, food appeal, and service quality as key factors in customer satisfaction. The gap is that most platforms focus only on general delivery apps. The Darb e-Link System addresses this by creating a simple, campus-based solution tailored to the needs of students and small vendors.

Development of a Web-Based Food Ordering System

Jiangmin (2024) introduced an online food ordering platform designed to improve efficiency for both restaurant staff and customers. The system features a user-friendly interface for browsing menus, placing orders, and selecting payment methods. Administrators can manage menus, generate reports, and process

transactions. The study highlights how personalized recommendations, seamless navigation, and data analytics contribute to enhanced customer satisfaction and support business decision-making in modern food services.

This system's strength is its user-friendly interface, multiple payment options, and report generation for admins. However, it is more suited for bigger food services. The Darb e-Link System fills this gap by being lightweight, affordable, and designed for small vendors like Darb, with features such as SMS notifications, COD, and pickup options.

The studies show that online food ordering systems improve speed, accuracy, and customer satisfaction, but most are made for big businesses. Local research also proves that students want convenience and reliable service, yet small vendors still struggle with delays and manual processes. The Darb e-Link System fills this gap by offering a simple, campus-based solution with SMS updates, flexible payments, and a pickup option to better serve both sellers and customers.

CHAPTER III
DESIGN AND METHODOLOGY

Introductory Paragraph

This chapter introduces the 4D Model: Discover, Design, Develop, and Deploy as the main approach for building the system. It explains how the group gathered information through interviews and observations, and how they used system tools like context diagrams, activity diagrams, and use case diagrams. It also shows the framework and diagrams that guided the creation of the Darb e-Link System.

Using the 4D Model provides a clear, step-by-step process that helps the team stay organized from discovery to deployment. It improves clarity, reduces errors and delays, and ensures that every phase of development is aligned with the overall project goals. These advantages make the 4D Model especially useful for the Darb e-Link System, as it guides the researchers in creating a well-structured and efficient solution (Harbinger Group, 2023).

Requirements Analysis

In order to determine the needed information in developing the web application, the researchers used the 4D Model. Developing the web application means building the system based on the design, which includes coding, setting up the database, adding features, and testing to make sure it works properly.

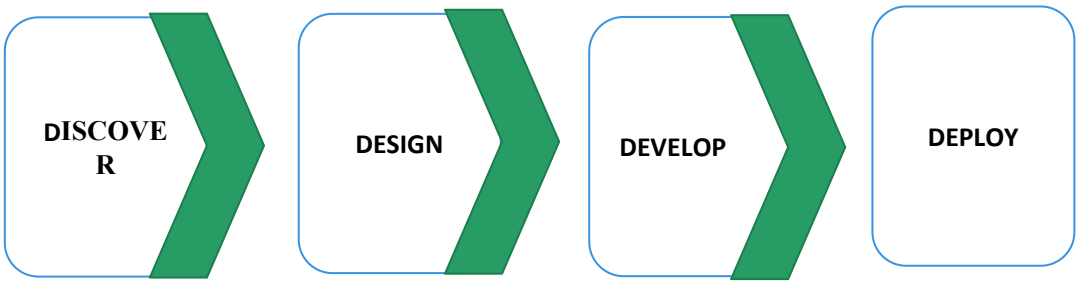


Figure 1. 4D Model

Discover Phase

The researchers gathered the data by interviewing the intended users on how to improve the current technical situation and address issues concerning delays.



Image 1 Interview and System Presentation

Image 1 shows the proponents interviewing the owner of Darb Food Stall while presenting a sample of the proposed online ordering system. The activity was part of the Discover phase, aimed at gathering insights and feedback to better understand the current workflow and improve the system design.

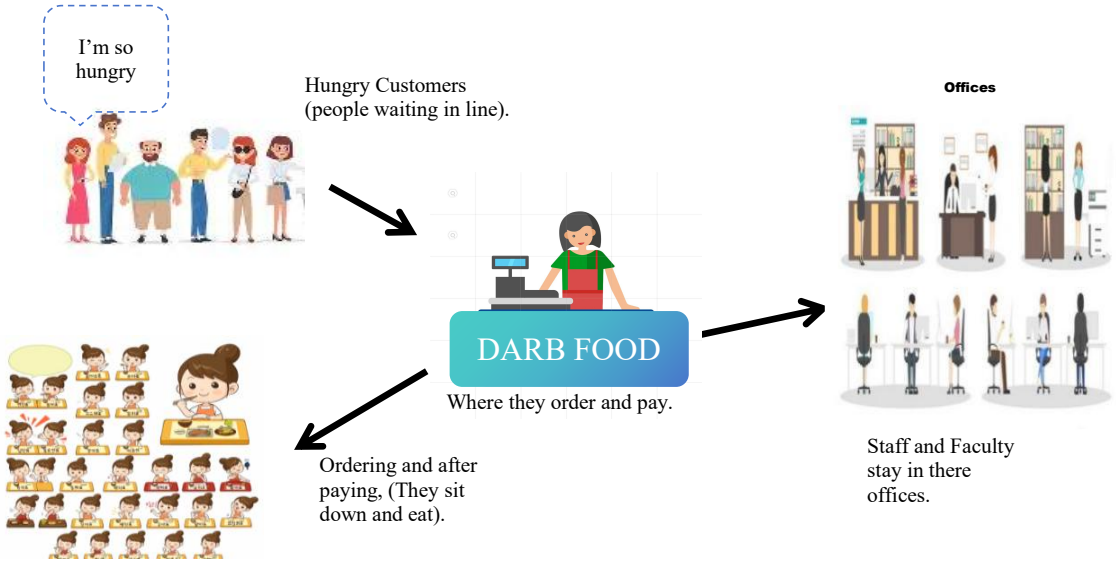


Figure 2. Current System Flow

Food consumers of the research environment will go to darb and buy foods of their choice while others prefers to order online using the social media.

Design Phase

In this phase the researchers start layouting the design of the system, taking in consideration the user friendliness of the web application.

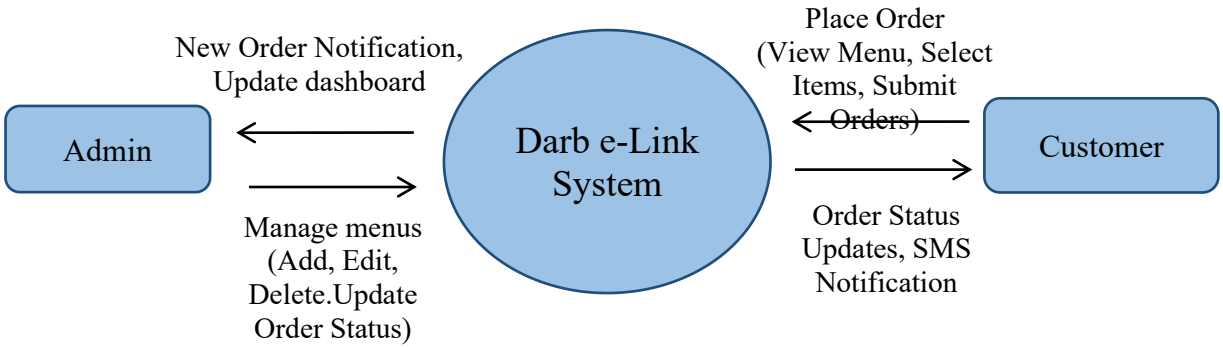


Figure 3. Context Diagram

The diagram shows the interaction of the two users with the proposed web application. The customer will automatically directed to the menu and does ot need to sign-up or login. The admin is responsible in managing the system, like adding menu, edit menu details, delete menu, view and update menu details and manage orders.The context diagram also features the system's integration with SMS gateways to automatically inform the admin upon the arrival of orders, being out for delivery, or delivered. This enhances real-time communication and simplifies order management.

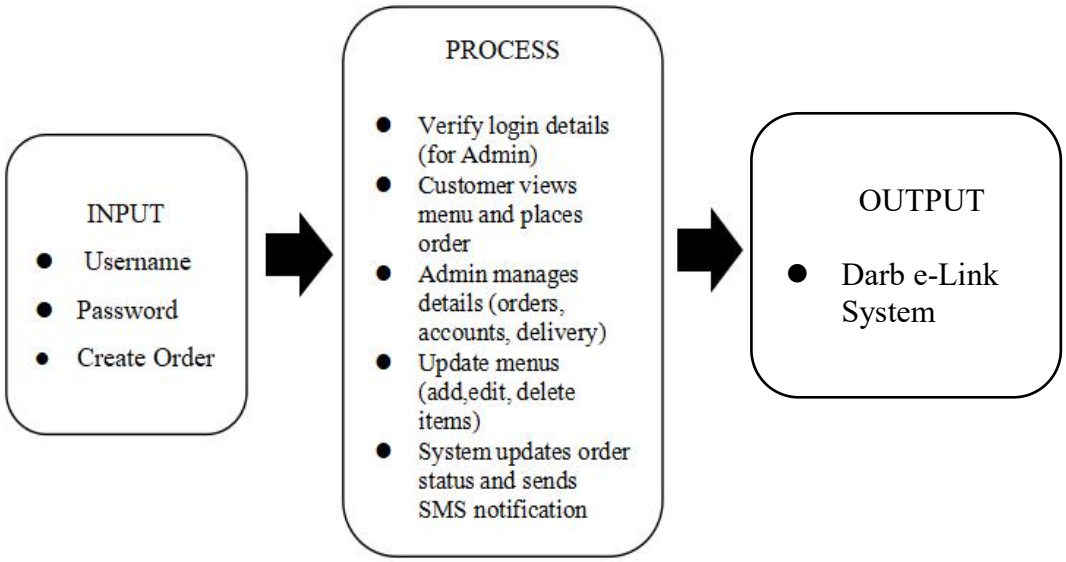


Figure 4. Conceptual Framework (IPO)s

Figure 4 shows the admin will input their username and password to log-in. The processes are (a) manage details (b) updates menus. Finally, the users can ordered food and pickup or delivery, view food menu and place order.

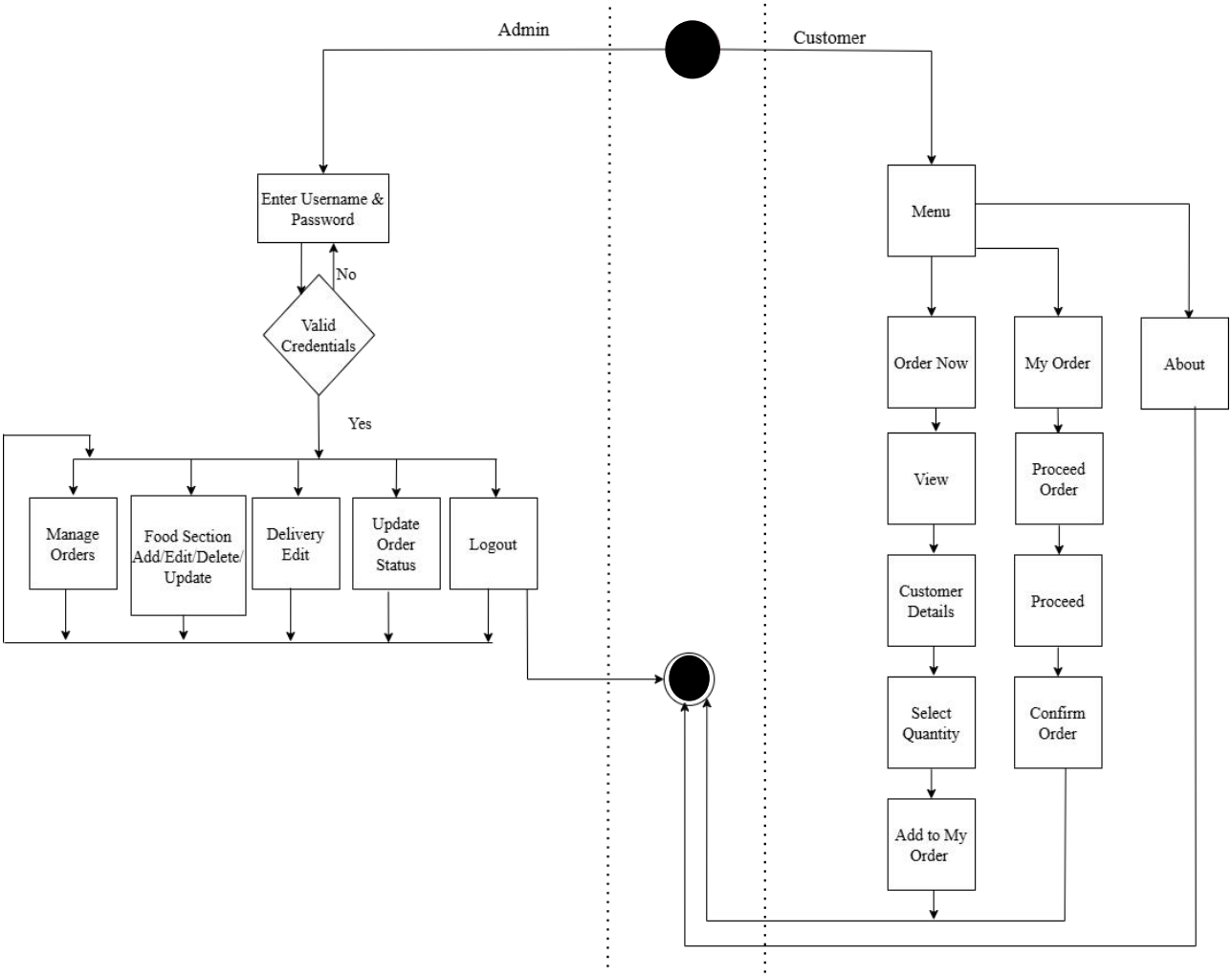


Figure 5. Activity Diagram (Admin/Customer)

The activity diagram shows the step-by-step flow of the Darb e-Link system for both Admin (left side) and Customer (right side). The process starts with a login where the system checks the account. After a successful login, the Admin can do different tasks such as managing orders, adding, editing, or deleting food items, viewing reports, editing account and delivery fee, updating the order status, and logging out. When the Admin changes the order status, the system also sends SMS

notifications to keep track of updates. On the other side, the Customer can open the menu, view food items, choose items with the desired quantity, and add them to the order. The Customer can then review the order, provide details, and confirm it for pick-up or delivery. After confirming, the system marks the order as in process and gives updates or instructions. This combined activity diagram clearly shows how both Admin and Customer use the system in a connected workflow.

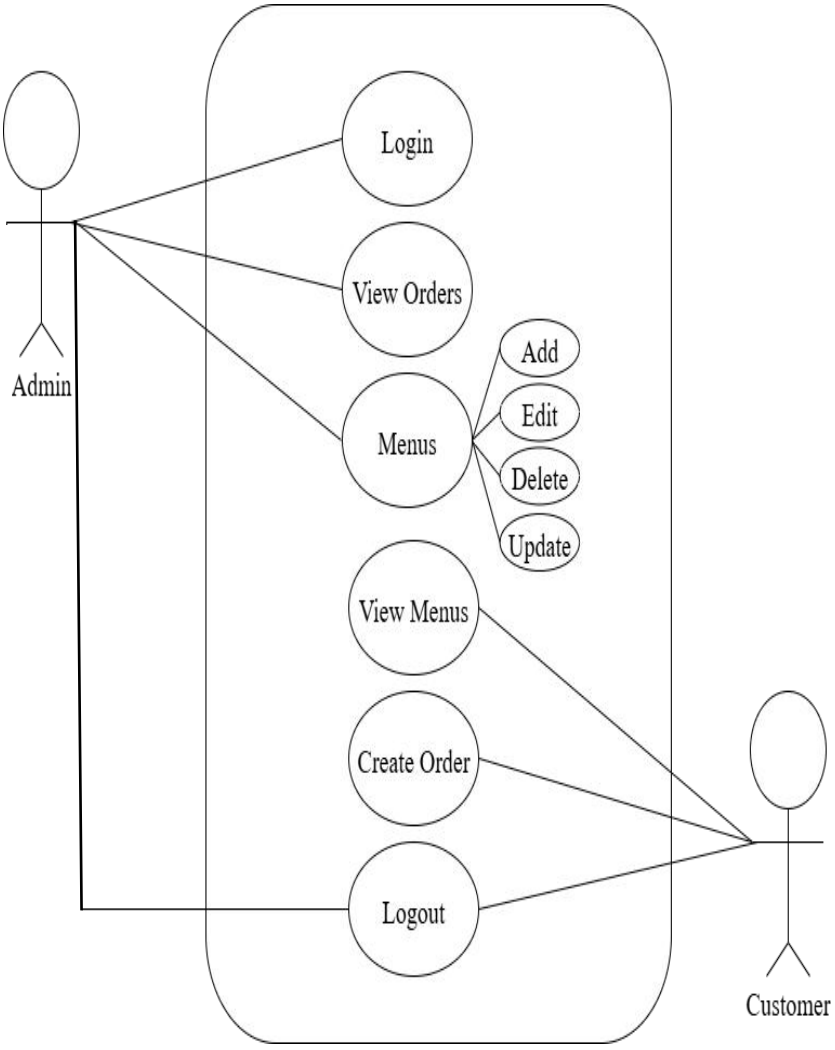


Figure 6. Use case Diagram

Figure 7 shows the interaction between the users (Admin and Customer) and the system. The Admin logs in and can view orders, manage menus (add, edit, delete, update), and log out. The Customer can view menus and create food orders.

Name	Menu
Description	System User Perform
Actor	Admin
Pre-Condition	Entry Product for the Customer's Order
Post-Condition	Receive the Customer's Order
Main Course	1. Add/Display Food 2. Receive Customer's Order 3. View Reports 4. Manage Order

Table 1. Use Case Description (Entry Food)

Name	Create Order
Description	System User Perform
Actor	Customer
Pre-Condition	Order Food
Post-Condition	Receive Order
Main Course	Create new order Receive Order

Table 2. Use Case Description (Create Order)

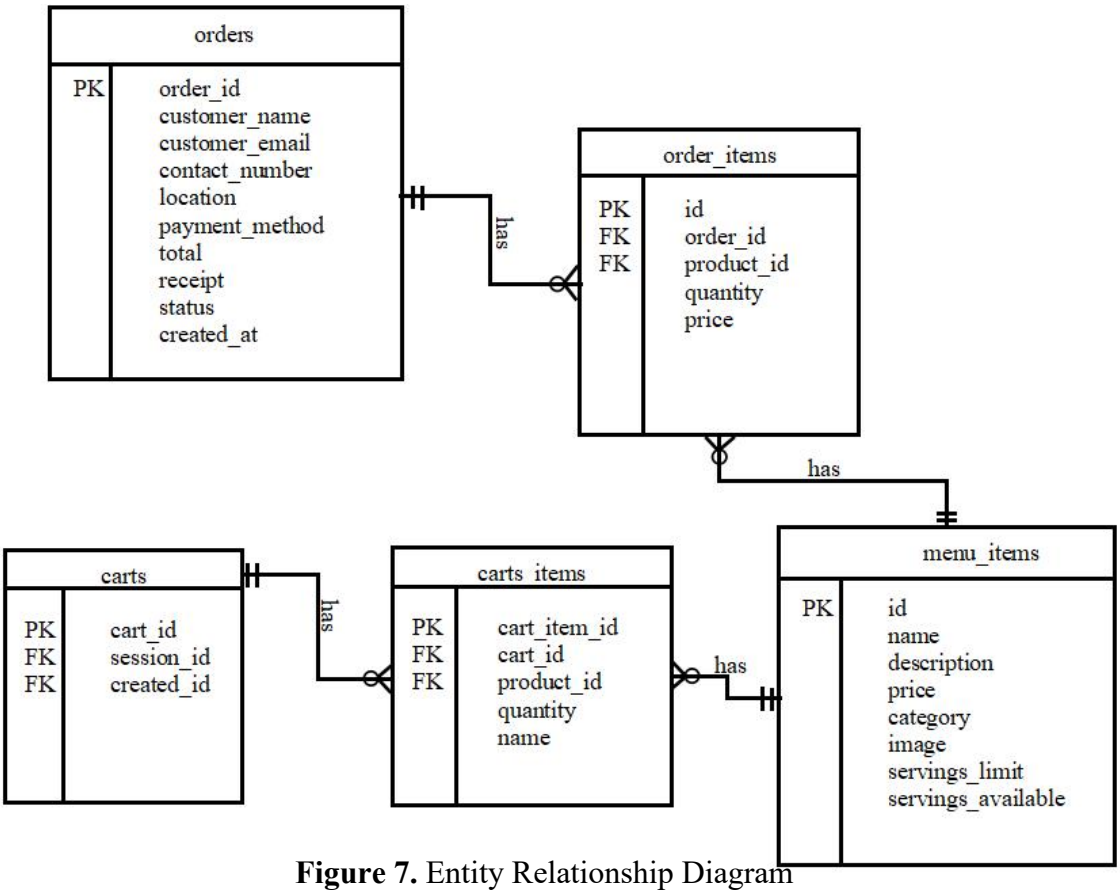


Figure 7. Entity Relationship Diagram

Figure 7 shows the connection of the different tables in the system. The orders table stores all customer order records and has a one-to-many relationship with the *order_items* table, as shown by the "has" notation. This means that one order can contain multiple order items, while each order item belongs to only one order.The *order_items* table acts as a linking entity between orders and *menu_items,* where each order item refers to a single menu item, but a menu item can appear in multiple order items. This relationship identifies the specific products included in each order. The *menu_items* table contains all food and product details and has one-to-many relationships with both *order_items* and *cart_items,* indicating that a menu item can be ordered multiple times and added to multiple carts.The *cart_items* table stores products added to the cart and has a many-to-one relationship with the carts table. This means each cart item belongs to one cart, while a cart can contain multiple

items.Overall, the **“**has**”** notation represents one-to-many relationships that allow the system to manage orders and carts efficiently.

Develop Phase

During this phase, the researchers built the Darb e-Link System by coding the website and setting up the database. This stage involves integrating the online menu, shopping cart, and order management features to ensure a fast and smooth food ordering process for users.

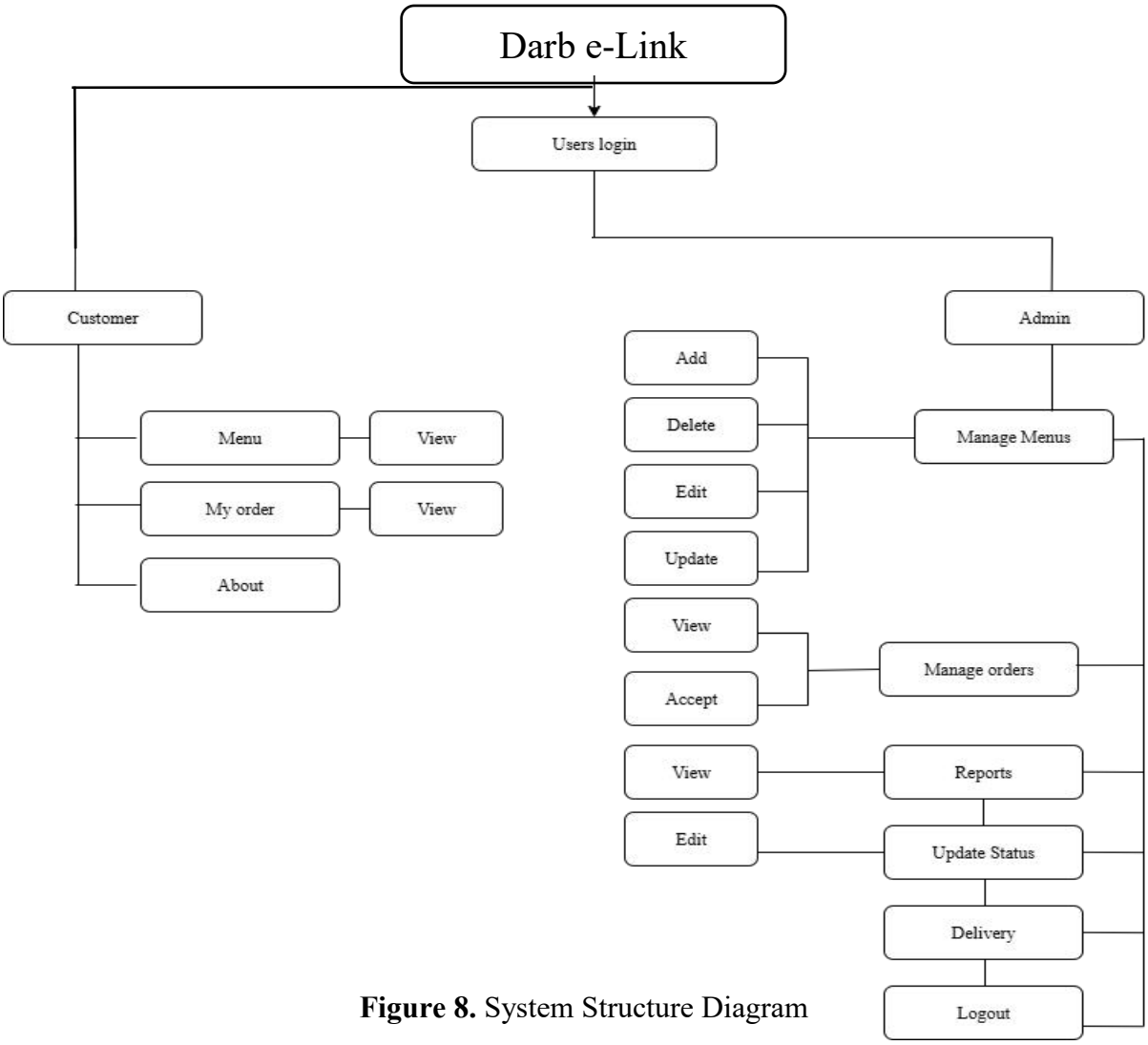


Figure 8. System Structure Diagram

Figure 8 shows the details of all the modules included in the proposed web application. The admin can log in to the system and manage its operations. The customer can access the system to view food menus and place orders for their desired

items. And update status demonstrates that the system is linked to an external SMS service to enable automated notifications.

Deploy Phase

In this stage, the researchers presented the final system, showing how the website, the database, and the ordering interface all work together. This phase ensures that the system is ready for use, providing a complete and functional platform for both the customers and the business.

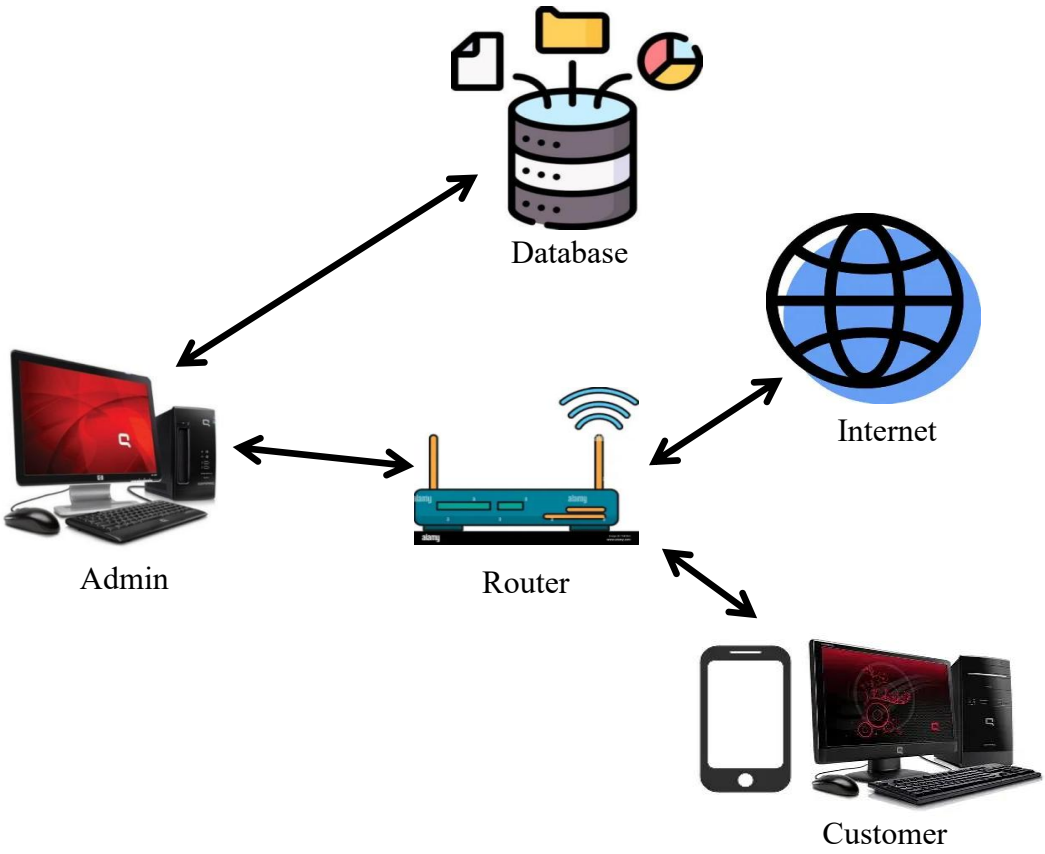


Figure 9. System Architecture Diagram

The diagram above illustrates how the following devices PC, smartphones, and router support the functional implementation of the web-application. The system is also connected to a database where all the data such as menu items, orders, and customer details are stored. The Admin manages the database through the system, making sure that orders, menus, and other records are updated properly.

CHAPTER IV
DEVELOPMENT AND TESTING

Description and Prototype

The prototype of Darb e-Link System has the following features. The Admin is responsible for managing the system through a secured dashboard, where they can add, edit, delete, and update menu items, monitor incoming orders, update order status, and send notifications to customers. The Customer, on the other hand, can access the online menu through a unique link, browse available food items, add them to cart, and place orders. Customers are also able to choose a payment method, either through GCash/Maya with receipt upload or Cash on Delivery (COD) where delivery personnel upload proof of payment.

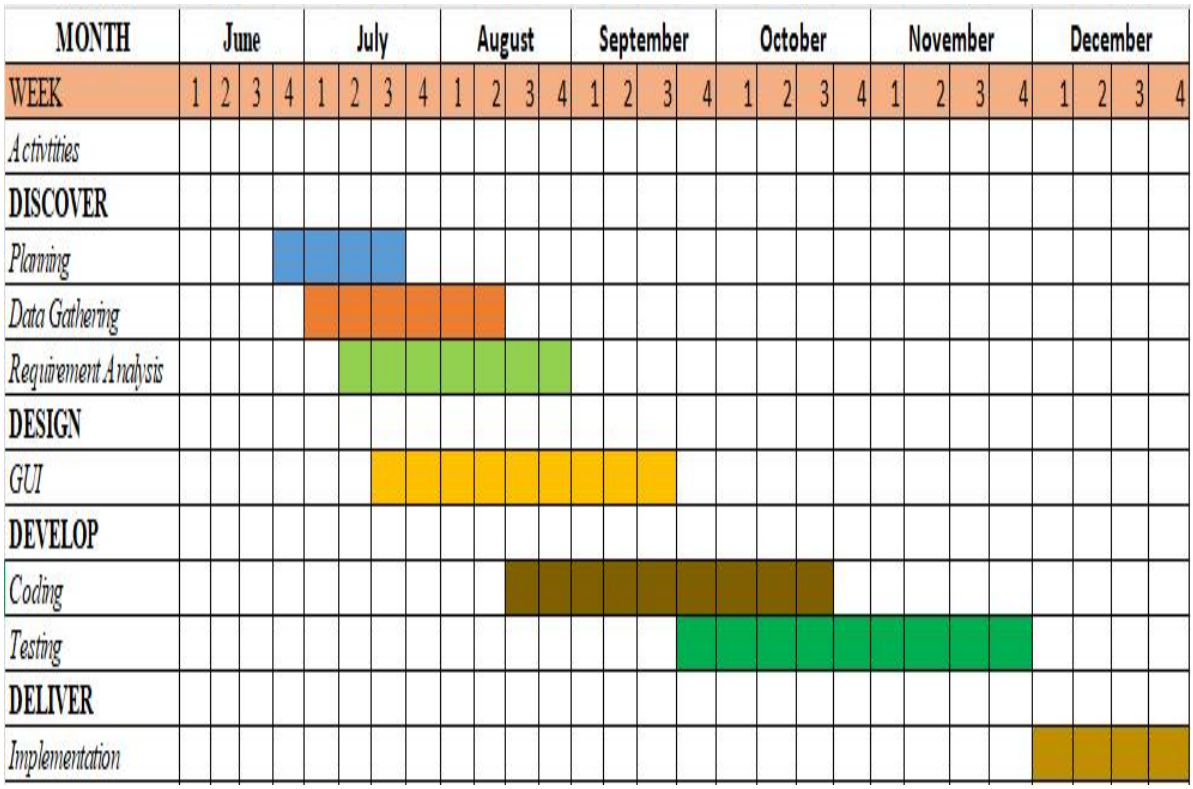


Table 3. Gantt Chart

The Gantt Chart represents the calendar of activities performed by the researchers. In the Discover phase, which includes the planning, data gathering, and

requirements analysis tasks, data was collected through interviews with intended users from the last week of June to August. In the Design phase, focusing on the user interface and system flow, work started in the third week of July and continued until the third week of September.

Development

The proponents used the 4D Model for the development of the Darb e-Link System which is suitable in creating a web-based application. In the development phase, the researchers ensured that the system functions according to the identified requirements and is useful to both the customers and the admin of the eatery. After the development of the system, a set of testing procedures was conducted to verify its functionality, usability, and reliability, making sure that it will operate to the best of its capabilities.

The following are the modules created in order to achieve the functionality of the proposed system.

Code

Menu Management Module

The Menu Management Module allows the administrator to add, update, and delete menu items and categories. The system processes actions using POST requests by checking conditions such as if *($action === 'add'), if ($action === 'update')*, and if *($action === 'delete').* When adding a menu item, the system validates the quantity and uploads the image using *$target_file = $target_dir* . basename*($_FILES["image"]["name"]);.*

Updating menu items allows the administrator to change item details, while deleting removes the item from the database. The module also allows category management, where categories can be added using if *($action === 'add')* and deleted

using if *($action === 'delete')*. This module ensures that menu information is properly managed and kept up to date.

```
<?php
$action = $_POST['action'] ?? '';
$id = $_POST['id'] ?? '';
$name = $_POST['name'] ?? '';
$price = $_POST['price'] ?? '';
$description = $_POST['description'] ?? '';
$category = $_POST['category'] ?? '';
$initialQuantity = $_POST['initial_quantity'] ?? '';
$currentQuantity = $_POST['quantity'] ?? '';
$image = $_FILES['image']['name'] ?? '';

if ($action === 'add') {
   if ($currentQuantity > $initialQuantity) {
      header('Location: menu.php?error=quantity_exceeds_initial');
      exit;   }

   // upload image
   $target_dir = "../assets/images/";
   $target_file = $target_dir . basename($_FILES["image"]["name"]);
move_uploaded_file($_FILES["image"]["tmp_name"], $target_file);
// create product
   createProduct($name, $price, $description, $category, $initialQuantity,
$currentQuantity, $image);
   header('Location: menu.php');
   exit; }

if ($action === 'update') {
   if ($currentQuantity > $initialQuantity) {
      header('Location: menu.php?error=quantity_exceeds_initial');
      exit;  }

  // if new image uploaded
 if (!empty($_FILES['image']['name'])) {
  $target_dir = "../assets/images/";
$target_file = $target_dir . basename($_FILES["image"]["name"]);
 move_uploaded_file($_FILES["image"]["tmp_name"], $target_file);
updateProduct($id, $name, $price, $description, $category, $initialQuantity,
$cur          rentQuantity, $_FILES['image']['name']);
 } else {
   updateProduct($id, $name, $price, $description, $category,
$initialQuantity,          $currentQuantity, null);
}header('Location: menu.php');
exit; }

if ($action === 'delete') {
```

```
    deleteProduct((int)$id);
    header('Location: menu.php');
    exit; }
$products = getAllProducts();
$categories = getAllCategories();
?>
<?php include '../includes/layout_admin.php'; ?>
<?php
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $action = $_POST['action'] ?? '';

  //   Add Category Function
    if ($action === 'add') {
        $name = trim($_POST['name'] ?? '');
        if ($name !== '') {
            addCategory($name);
        }
        header('Location: /admin/categories.php');
        exit;  }

//   Delete Category Function
    if ($action === 'delete') {
        $id = $_POST['id'] ?? '';
        if ($id !== '') {
            deleteCategory($id);
        }
        header('Location: /admin/categories.php');
        exit;}
}?>
```

Notification Module

The Notification Module sends SMS updates to customers to inform them about the status of their orders. The system creates the SMS message based on the order status using a switch (*$status*) statement, where each case such as pending, confirm, out for delivery, ready to pickup, picked up, and delivered has its own message that includes the customer's name and total amount.

After generating the message, the module checks if the message is not empty using *if (!empty($sms_message))*. The customer's contact number is then formatted using preg_replace() and converted to the correct format when it starts with 09. Finally, the SMS is sent to the customer using the httpsms API, ensuring that

customers receive timely updates about their order progress.

```
        // Customize message per status
           switch ($status) {
              case 'pending':
                 $sms_message = "Order Received - Hi {$customer_name}, your order
is               being processed.\nTotal: {$total_text}\n- Darb Food Ordering";
                 break;
              case 'confirm':
                 $sms_message = "Order Confirmed - Hi {$customer_name}, your
order              has been confirmed.\nTotal: {$total_text}\n- Darb Food Ordering";
                 break;
              case 'out for delivery':
                 $sms_message = "Order Out for Delivery - Hi {$customer_name},
your               order is now out for delivery.\nTotal: {$total_text}\n- Darb Food
Ordering";
                 break;
              case 'ready to pickup':
                 $sms_message = "Order Ready for Pickup - Hi {$customer_name},
your               order is ready for pickup!\nTotal: {$total_text}\n- Darb Food
Ordering";
                 break;
              case 'picked up':
                 $sms_message = "Order Picked Up - Thank you {$customer_name}!
                 Enjoy your meal.\nTotal: {$total_text}\n- Darb Food Ordering";
                 break;
              case 'delivered':
                 $sms_message = "Order Delivered - Hi {$customer_name}, your
order               has been delivered!\nTotal: {$total_text}\n- Darb Food Ordering";
break;   }

        // Send SMS using httpsms API
           if (!empty($sms_message)) {
              $sms_api_url = 'https://api.httpsms.com/v1/messages/send';
              $sms_api_key =
        'uk_w2adRlCdJPjmFj_nLMEu0cEUbp16RHeHHEuu7J4BLXLE14dz4tbxi_kD
vA                 mP_uf4';
              // Format contact number
              $contact_number = preg_replace('/[^0-9+]/', '', $order['contact']);
              if (strpos($contact_number, '09') === 0) {
                 $contact_number = '+63' . substr($contact_number, 1);
              }
              $sms_data = [
                 'from' => '+639940996035',
                 'to' => $contact_number,
                 'content' => $sms_message,   ];
```

Alpha Testing

Alpha testing is performed to check whether the features of the Darb e-Link System work correctly according to the intended design. This testing is done to ensure that each function such as logging in, managing products, placing orders, updating statuses, and viewing menus responds properly when used by administrators and customers.

The purpose of this test is to confirm that all system features operate smoothly, display accurate outputs, and do not produce errors during normal use. By documenting each test case, the team can verify which functions are working correctly and identify parts of the system that may still need adjustment before final deployment.

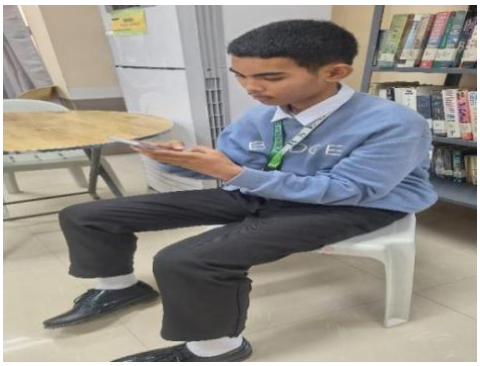


Image 2



Image 3



Image 4

Image 4 shows the Owner of Darb food stall, and image 2 and 3 shows the selected customers of onyx. The researchers let the user operate and explore the the system. During testing the researchers receives some comments and suggestions from the owner is about the name of the stall to change from Onyx to Darb.

Compatibility Checking

Hardware

Desktop Computer – any type of processor such as Intel Core i3, i5, i7, Celeron, Pentium, or equivalent with at least 4GB RAM and 300GB storage.

Mobile Phone – any modern smartphone (Android or iOS) capable of running updated web browsers.

Router – any standard router or Wi-Fi access point to connect devices to the internet.

Software

Browser – the system is accessible through commonly used browsers such as Google Chrome, Mozilla Firefox, or Microsoft Edge.

Operating System – compatible with Windows 10 and higher, Android, and iOS devices.

Implementation Plan

Project Phase	System Activities	Persons Involved	Duration
Approval and Preparation	Install XAMPP, MySQL, PHP, and VS Code. Set up the system and upload it online using InfinityFree (domain and database setup).	System Administrator	2 weeks
Installation / Setup	Test the online system including customer ordering, admin functions, and SMS notifications.	System Administrator, Customer	1 week & 3 days
Orientation and Demonstration	Train users on how to use the system, manage the menu, update orders, and handle customer transactions.	System Administrator, Customer	4 days

Table 3. Implementation

Table 3 shows the implementation of the Darb e-Link System after it was uploaded online using InfinityFree. The whole process lasted 5 weeks. The first phase, Approval and Preparation (1 week), included installing the needed software and setting up the domain and database for the system. The second phase, Installation/Setup (1 day), focused on testing the online system to make sure customer orders, admin features, and SMS notifications were working properly. The final phase, Orientation and Demonstration (1 day), involved giving users basic training on how to use the admin dashboard, update the menu, manage orders, and handle customer transactions.

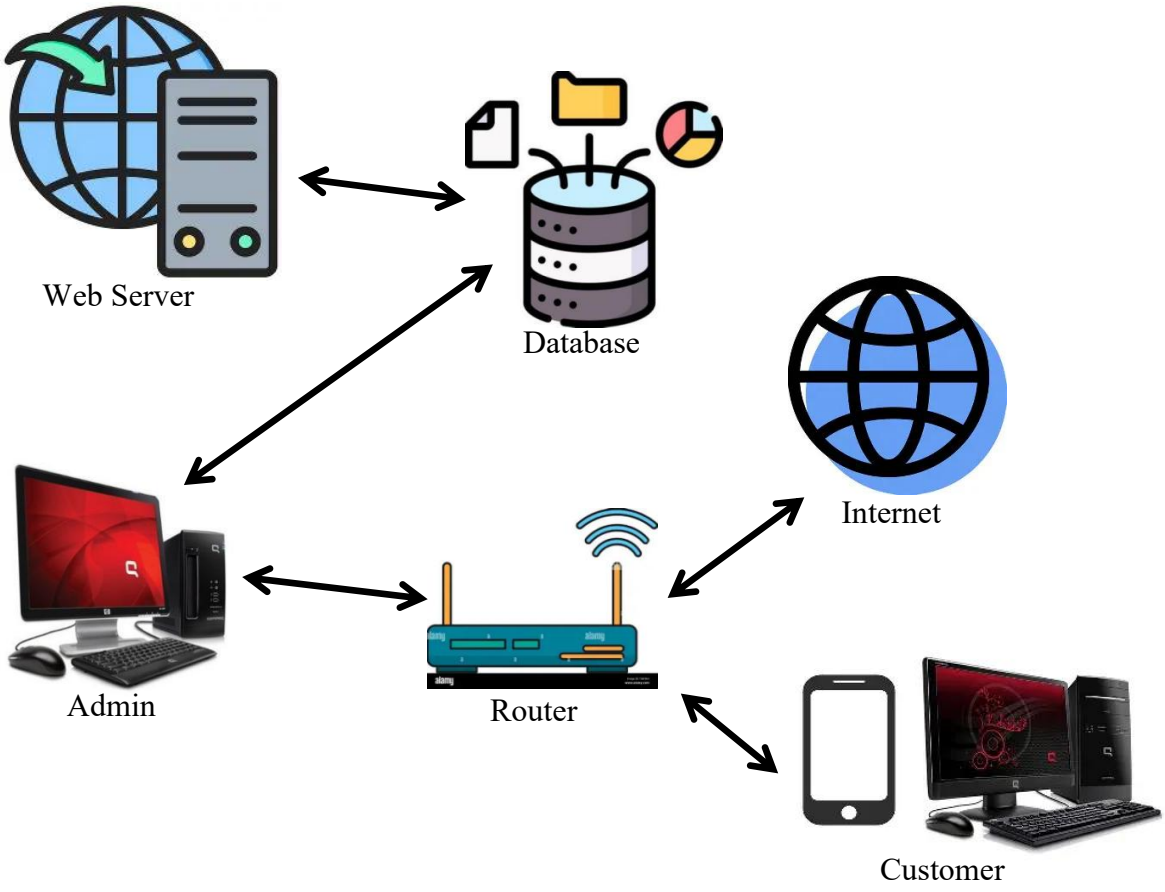


Figure 10. Deployment Diagram

Figure 10 shows the deployment diagram of the Darb e-Link System. Users (customers and administrators) can access the system using desktop computers or

mobile phones through a web browser with an internet connection. On the server side, the system is hosted on an InfinityFree web hosting server that supports PHP and is connected to a MySQL database. The web server handles order processing, menu management, and order status updates, while the database stores customer information, menu items, orders, and payment records. Communication between users and the system occurs through the internet.

CHAPTER V

IMPLEMENTATION RESULTS, DISCUSSION, CONCLUSION AND RECOMMENDATION

Implementation Result and Discussion

This chapter presents the results of the Darb e-Link System implementation. It includes screenshots and descriptions of the system's main features and user interfaces. The discussion shows how each part of the system works to provide an easy and efficient way for customers and the admin to manage online food orders.

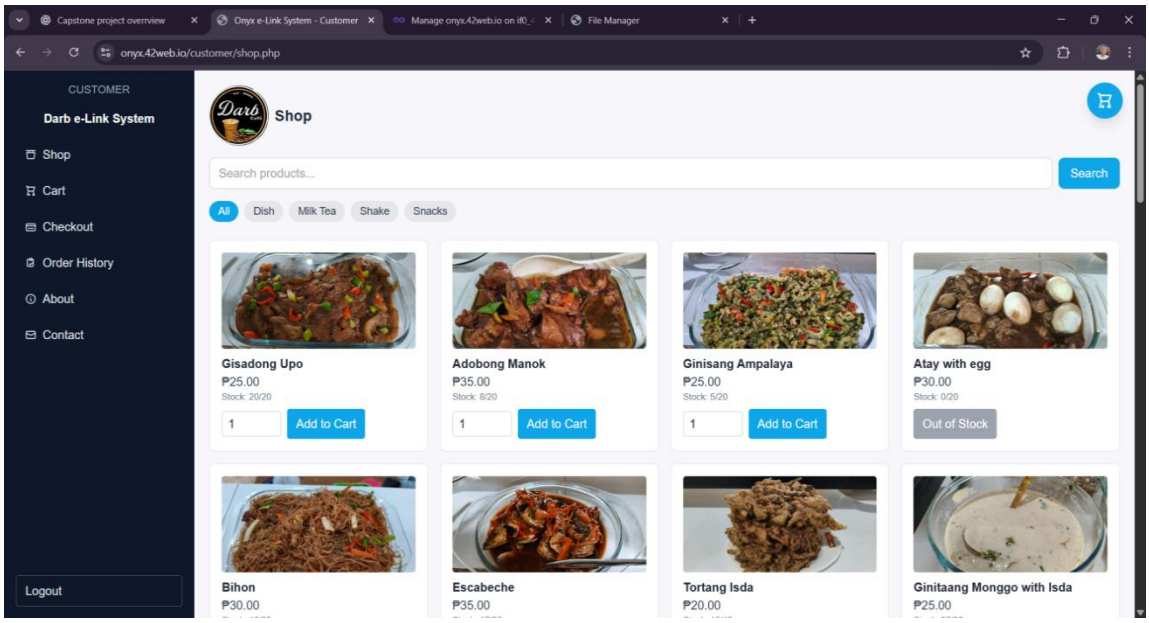


Image 5. Menu

Image 5 shows the food menu. It allows the user to order food, choose a category to browse, and use the search bar to find what they want. The customer can also select how many items or quantity they want to order.

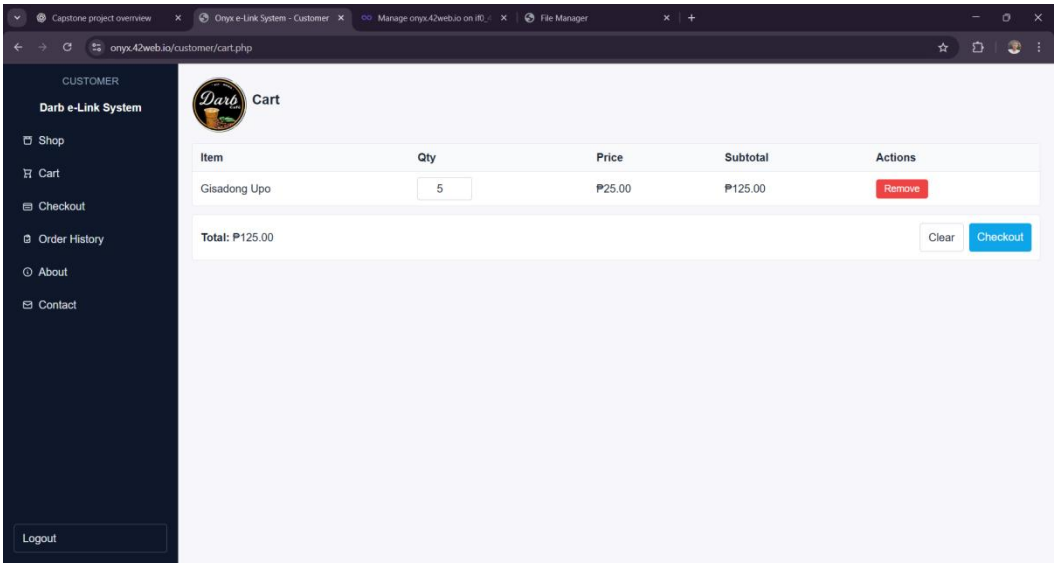


Image 6. Cart

Image 6 shows the cart page. Here, the customer can see all the items added to the cart. They can also increase or decrease the quantity of each item before checking out.

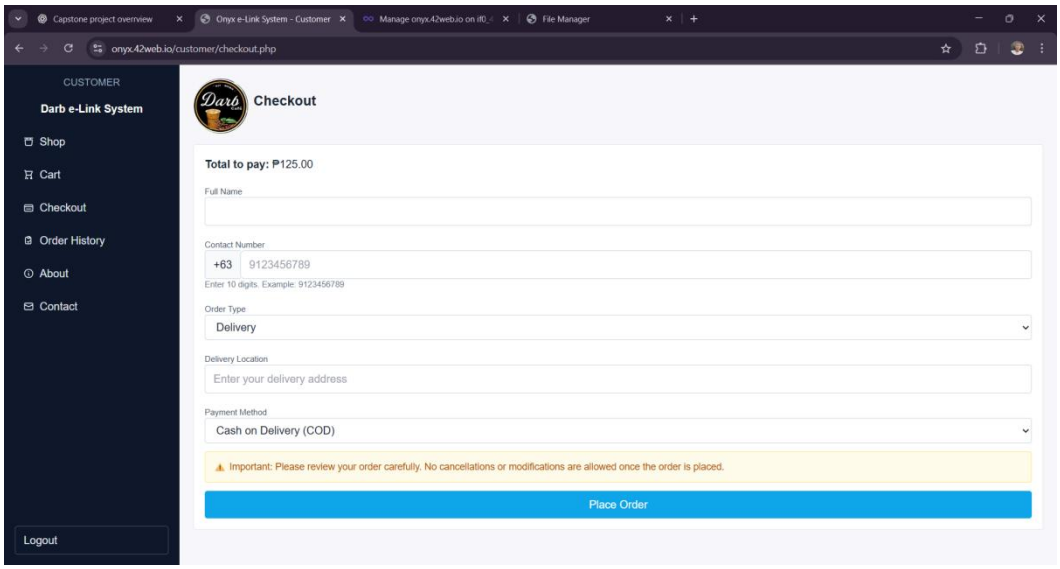


Image 7. Checkout

Image 7 shows the checkout page. Before placing the order, the customer must fill in their name, contact number, delivery type (pickup or delivery), location, and payment method (GCash, Maya, or Cash on Delivery). A note also reminds customers to review their order carefully since changes or cancellations are not allowed after

placing it.

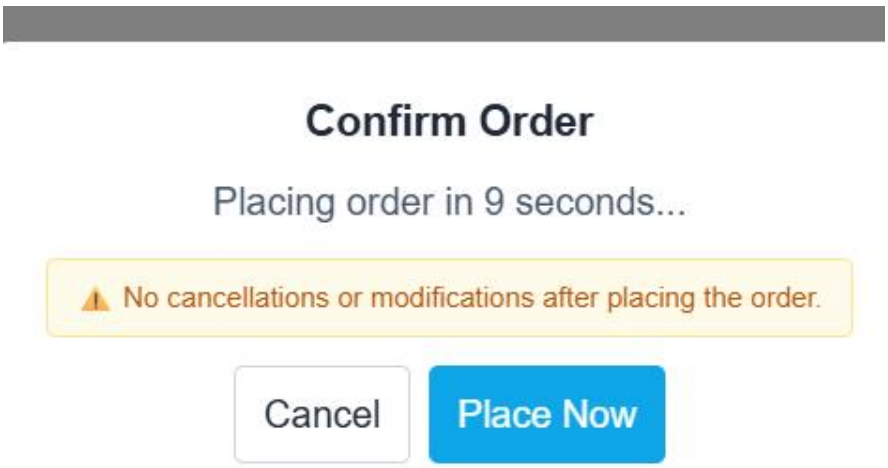


Image 8. Confirm Order

Image 8 shows a 10-second timer. The customer can still cancel the order within this time if they want to add or change something. After the timer ends, the order is automatically placed and can no longer be canceled.

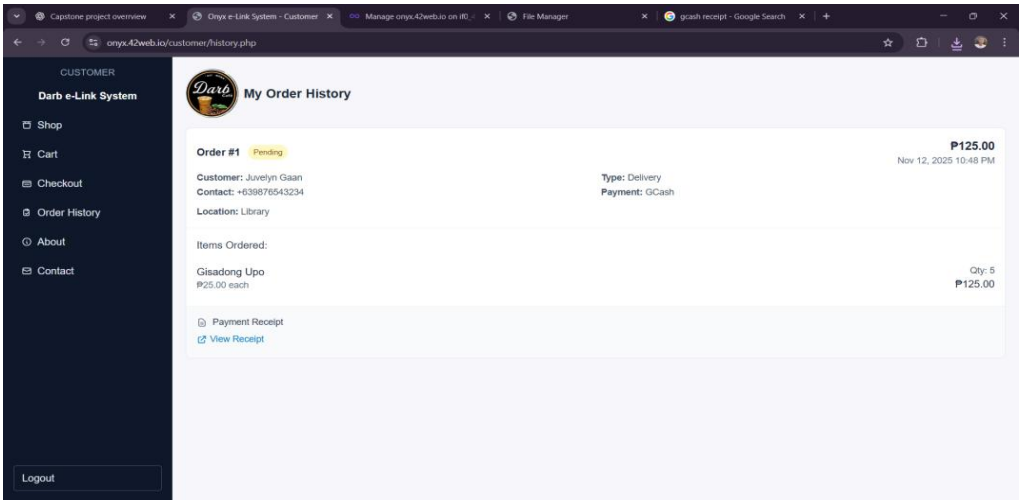


Image 9. Order History

Image 9 shows the order history page where the customer can see all the items they have previously ordered.

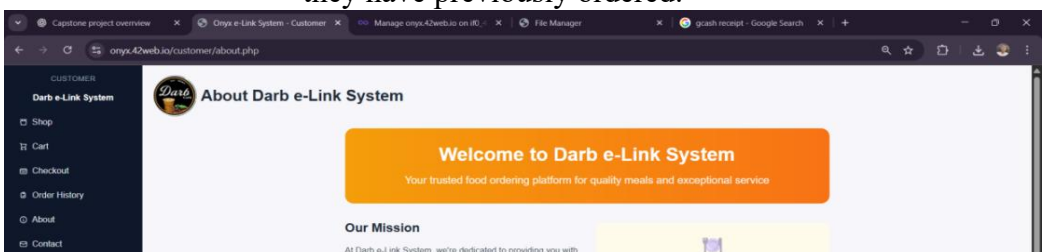


Image 10. About

Image 10 shows the proposed Darb e-Link System and its main features. The system is easy to use. Customers can place, edit, or cancel orders and view their past orders. Admin can manage the menu by adding or removing items. Both customers and admin can track order status. The system makes ordering and managing food faster and simpler.

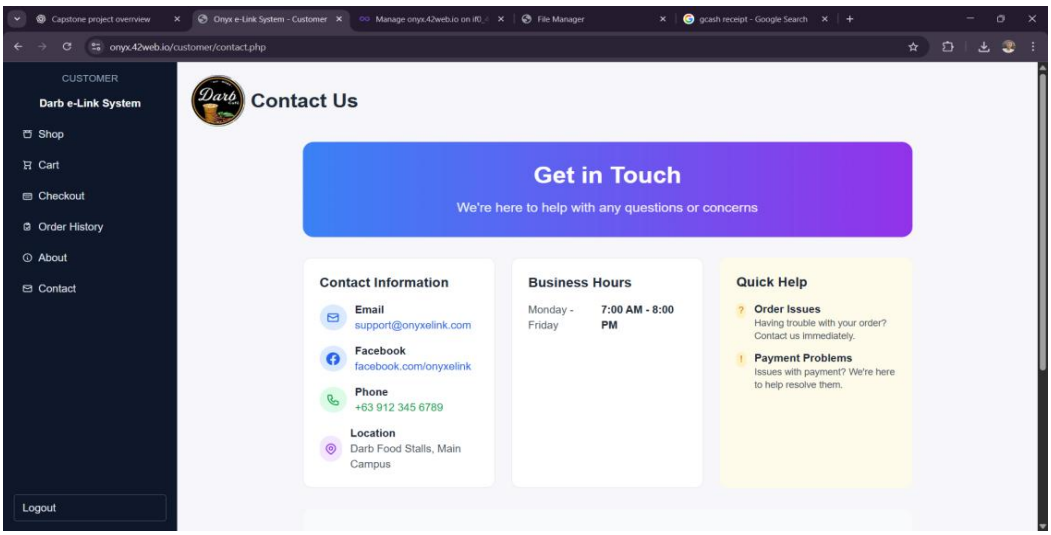


Image 11. Contact Us

Image 11 shows on how customers can contact to the owner of Darb e-Link System.

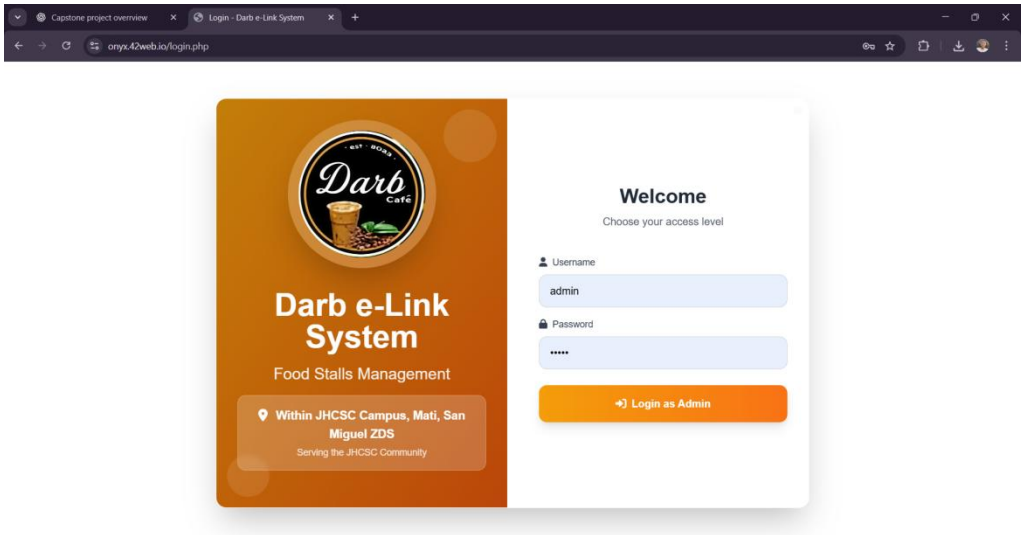


Image 12. Login

Image 12 shows the Admin Login page the login system of the Darb e-Link System relies on username, password, and user role data to ensure secure authentication and proper access control.

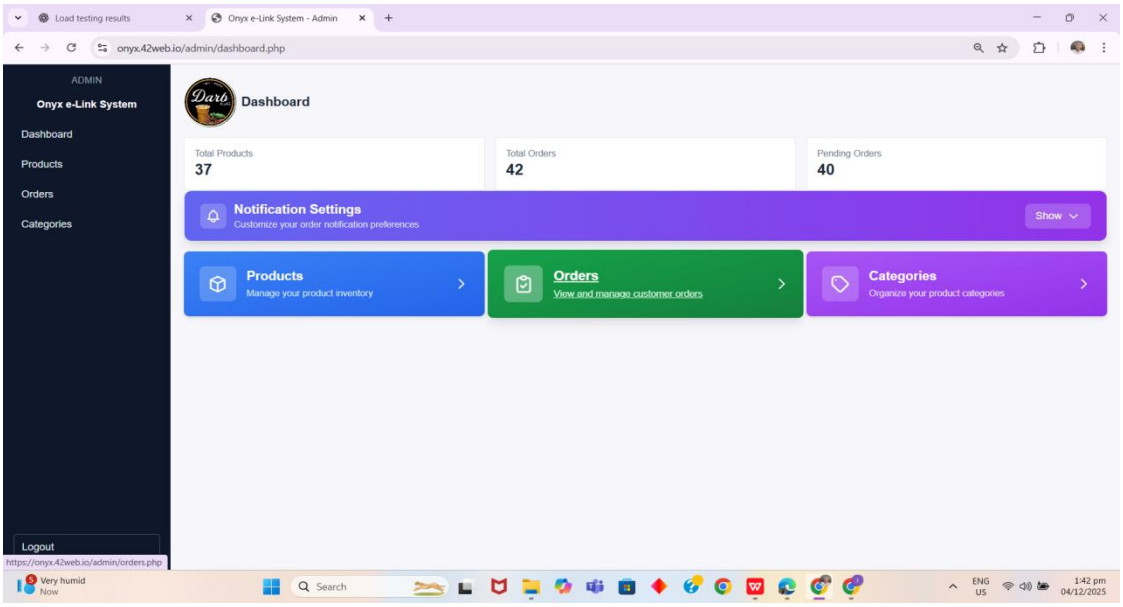


Image 13. Dashboard

Image 13 shows the admin dashboard, admin can show how many total products, total orders, and total of pending orders. The Admin can also customize the sound of the notification.

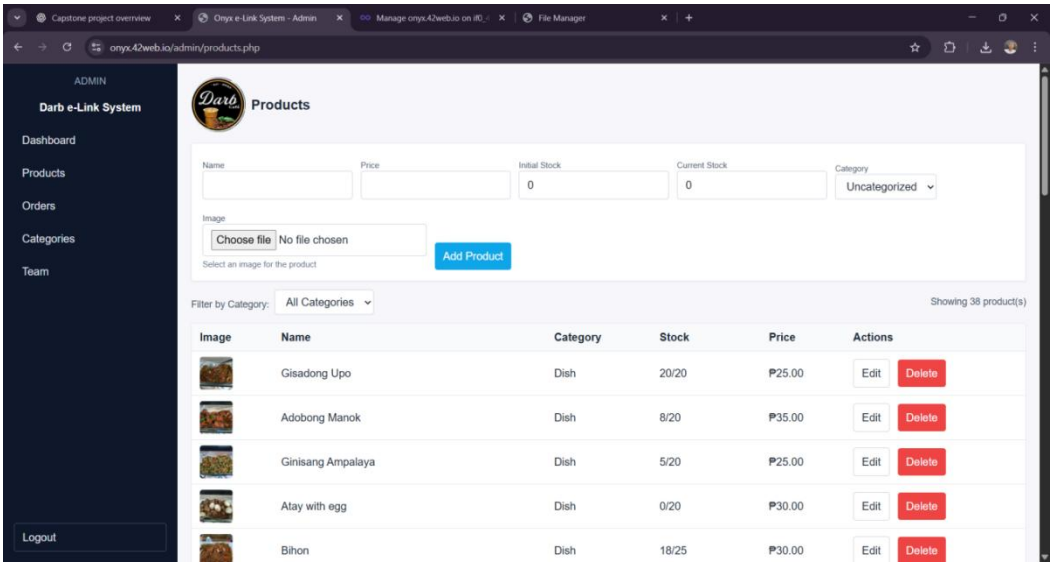


Image 14. Products

Image 14 shows how the admin can add products to the system. The admin can see all the products they have created below. They can also choose which category to display each product in. This makes it easy for the admin to organize and manage the products.

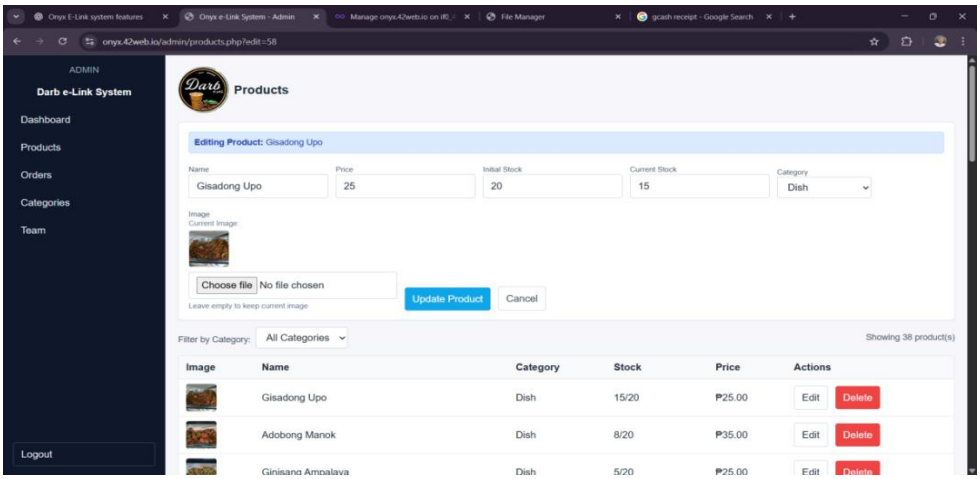


Image 15. Products

Image 15 shows the editing products when clicking the edit button of the created products.

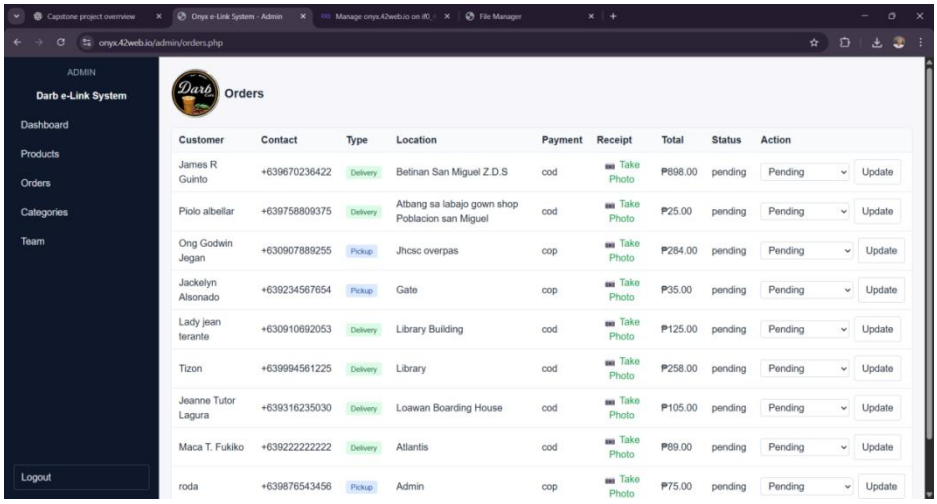


Image 16. Orders

Image 16 shows all the orders in the system. The admin can view each order's receipt and the payment method chosen by the customer. The admin can also update the order status to (Confirmed, Out for Delivery or Delivered) or (Confirmed, Ready for Pick up or Picked up). This helps the admin manage and track all customer orders easily.

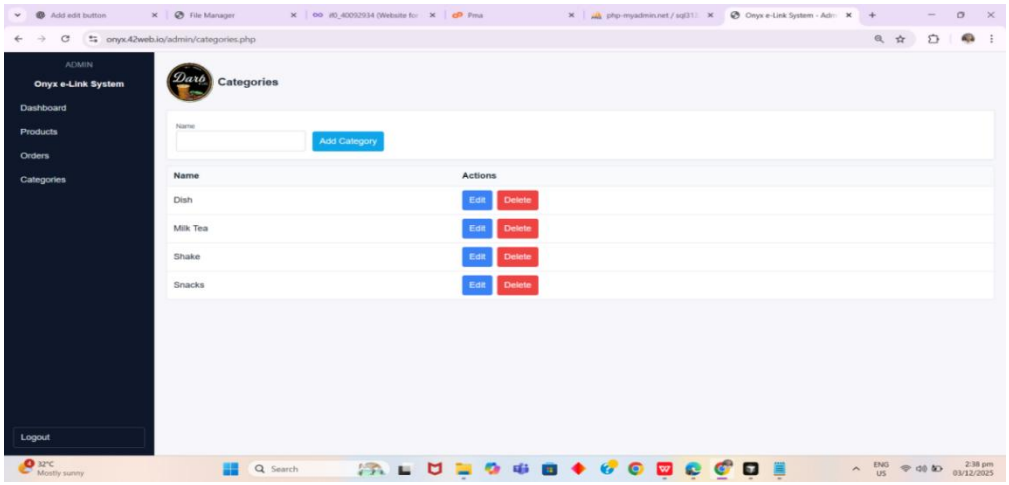


Image 17. Categories

Image 17 show the creating of categories for the system.

1. Functionality Test Cases

Purpose: Based on the functionality testing, all system features worked correctly as intended (see appemdix A) Both admin and customer functions such as login, viewing the menu, placing orders, uploading payment proof, managing orders, and receiving SMS notifications performed successfully. The ordering process for delivery and pickup, including cancellation and auto-submit features, functioned properly. The system also handled multiple users at the same time without errors or performance issues, indicating that it is stable, reliable, and ready for use.

2. Reliability Testing

Purpose: Reliability testing is conducted to check if the Darb e-Link System can perform its functions (see appendix A) correctly and consistently, even when unexpected problems occur. This test observes how the system behaves during repeated actions, sudden interruptions, duplicate submissions, and long periods of use. The purpose of this testing is to make sure the system remains dependable, does not lose data, and continues to operate normally under different situations.

3 . Usability Testing

Purpose: Usability testing is carried out to evaluate how easy the Darb e-Link System is to understand (see appemdix A) and use, especially for new users. This test focuses on navigation clarity, visibility of buttons, readability of error messages, and how quickly users can complete tasks. The purpose of this testing is to ensure that the system provides a smooth and user-friendly experience for all types of users.

4. Efficiency Testing

Purpose: Efficiency testing was conducted to evaluate the speed and responsiveness of the Darb e-Link System under normal and moderate usage conditions. This testing includes (see appemdix A) checking page loading time, dashboard responsiveness,

resource usage, and system performance during multiple user access. Since the system is deployed on an free hosting platform, minor delays may occur depending on internet speed and server availability. However, the purpose of this testing is to confirm that the system can still operate efficiently, remain usable, and perform its functions properly without significant delays or system failure.

5. Compatibility Testing

Purpose: Compatibility testing was conducted to check whether the Darb e-Link System works properly (see appemdix A) across selected browsers, devices, screen resolutions, and operating systems. This testing focuses only on the environments that were applied during the testing phase. The purpose of this testing is to ensure that the system remains functional, consistent, and usable when accessed using the tested platforms.

6. Load Testing Procedure

Purpose: Load testing is conducted to evaluate the performance and stability of the Darb e-Link System when (see appemdix A) many users access the system at the same time. The purpose of this test is to ensure that the system can handle multiple customers browsing the menu, placing orders, and viewing order status without slowing down or crashing.

During the test, several users were allowed to use the system simultaneously to simulate real situations such as peak hours in the campus. The results helped determine if the system can continue to operate smoothly while processing multiple orders and admin activities at once.

7. Stress Testing Procedure

Purpose: Stress testing is conducted to see how the Darb e-Link System performs under very heavy usage. (see appemdix A) This test checks what happens when many

customers place orders at the same time, when orders are submitted repeatedly, or when the server experiences high traffic during peak hours.

The purpose of this testing is to identify the limits of the system, observe possible slowdowns or errors, and determine which parts of the system may need improvement. The results help ensure that even during busy periods, the system can still operate and recover properly.

8. Beta Testing

Purpose: Beta testing was conducted to check the usability and performance of the Darb e-Link System (see appemdix A) using actual users such as students, staff, and faculty. The testing focused on the system's main features to ensure that it works properly and is easy to use in real food ordering situations.

The beta testing of the Darb e-Link System was conducted with 55 participants to evaluate the system's performance, usability, and reliability in real-world use. After the testing phase, a Survey Questionnaire was distributed using a Likert scale with the options Strongly Disagree, Disagree, Neutral, Agree, and Strongly Agree to collect feedback on ease of use, interface design, system responsiveness, and overall user satisfaction. The results were used to further improve and refine the system before its official implementation.

Survey Questionnaire of the Darb e-Link System

This survey is conducted for academic and research purposes only as part of the capstone project titled "Darb e-Link System". All responses will be treated with confidentiality and used solely for educational evaluation and improvement of the system.

The questionnaire used in this study is provided in **Appendix D.**

Table 3
Darb e-Link System

5 – Strongly 4 – Agree 3 – Neutral 2 – Disagree 1 – Strongly Disagree

ALL	5	4	3	2	1	TWP	WM	Descr.	Rank
1. The Darb e-Link System is easy to learn and understand.	25	24	6	0	0	55	4.35	SA	9
2. I can navigate the Darb e-Link System without any difficulty.	32	19	4	0	0	55	4.51	SA	6
3. The layout and functions of the system are simple and user-friendly.	38	13	4	0	0	55	4.62	SA	3
4. I can easily find the food items or options I need in the system.	38	13	4	0	0	55	4.62	SA	3
5. The ordering process in the Darb e-Link System is straightforward and clear.	35	14	6	0	0	55	4.53	SA	5
6. I can complete my food order with minimal steps or effort.	35	12	8	0	0	55	4.49	SA	7
7. The instructions and labels in the system are easy to understand.	35	16	4	0	0	55	4.56	SA	4
8. The system responds quickly when I click or submit actions.	34	20	1	0	0	55	4.60	SA	2
9. I can use the Darb e-Link System without needing help from others.	40	13	2	0	0	55	4.69	SA	1
10. The system works smoothly and consistently across both mobile and desktop devices.	41	10	4	0	0	55	4.67	SA	2
					Total		**4.56**	**SA**	

Overall, the 55 respondent results show that the Darb e-Link System is highly acceptable and easy to use based on the total weighted mean of 4.56, which is interpreted as Strongly Agree (SA). This means that most users find the system easy to learn, simple to navigate, and user-friendly. Respondents agreed that ordering food

is clear, requires minimal effort, and can be done without help from others. The system also responds quickly and works smoothly on both mobile and desktop devices. In general, the findings indicate that the Darb e-Link System is effective, efficient, and suitable for users, making it reliable for food ordering purposes in this study.

Table 4
Darb e-Link System

5 – Strongly 4 – Agree 3 – Neutral 2 – Disagree 1 – Strongly Disagree

Student	5	4	3	2	1	TWP	WM	Descr.	Rank
1. The Darb e-Link System is easy to learn and understand.	8	6	8	0	0	22	4.00	A	10
2. I can navigate the Darb e-Link System without any difficulty.	9	8	5	0	0	22	4.18	A	9
3. The layout and functions of the system are simple and user-friendly.	11	9	2	0	0	22	4.41	SA	6
4. I can easily find the food items or options I need in the system.	12	8	2	0	0	22	4.50	SA	3
5. The ordering process in the Darb e-Link System is straightforward and clear.	11	8	3	0	0	22	4.36	SA	7
6. I can complete my food order with minimal steps or effort.	10	9	3	0	0	22	4.23	SA	8
7. The instructions and labels in the system are easy to understand.	11	9	2	0	0	22	4.41	SA	6
8. The system responds quickly when I click or submit actions.	11	10	1	0	0	22	4.45	SA	4

9. I can use the Darb e-Link System without needing help from others.	13	8	1	0	0	22	4.59	SA	1
10. The system works smoothly and consistently across both mobile and desktop devices.	12	8	2	0	0	22	4.45	SA	4
					Total		**4.46**	**SA**	

Overall, the 22 student respondents strongly agreed with the usability of the Darb e-Link System, as shown by the total weighted mean of 4.46, interpreted as Strongly Agree (SA). This indicates that students find the system easy to learn, simple to navigate, and user-friendly. They can easily find food items, place orders with minimal effort, and understand the instructions without needing help from others. The system is also fast in responding and works smoothly on both mobile and desktop devices. In general, the results show that the Darb e-Link System is effective and convenient for student users, making it suitable for use in food ordering transactions.

Table 5
Darb e-Link System

5 – Strongly 4 – Agree 3 – Neutral 2 – Disagree 1 – Strongly Disagree

Staff	5	4	3	2	1	TWP	WM	Descr.	Rank
1. The Darb e-Link System is easy to learn and understand.	8	6	3	0	0	17	4.35	SA	10
2. I can navigate the Darb e-Link System without any difficulty.	9	6	2	0	0	17	4.51	SA	8
3. The layout and functions of the system are simple and user-friendly.	10	5	2	0	0	17	4.62	SA	3.5
4. I can easily find	1	5	2	0	0	17	4.62	SA	3.5

the food items or options I need in the system.	0								
5. The ordering process in the Darb e-Link System is straightforward and clear.	9	5	3	0	0	17	4.53	SA	7
6. I can complete my food order with minimal steps or effort.	9	4	4	0	0	17	4.49	SA	9
7. The instructions and labels in the system are easy to understand.	9	6	2	0	0	17	4.56	SA	6
8. The system responds quickly when I click or submit actions.	9	7	1	0	0	17	4.60	SA	5
9. I can use the Darb e-Link System without needing help from others.	11	4	2	0	0	17	4.69	SA	1
10. The system works smoothly and consistently across both mobile and desktop devices.	11	4	2	0	0	17	4.67	SA	2
					Total		**4.564**	**SA**	

Overall, the staff respondents strongly agreed on the usability of the Darb e-Link System, as shown by the total weighted mean of 4.564, interpreted as Strongly Agree (SA). This means that the staff find the system easy to learn and navigate, with a simple and user-friendly layout. They can easily find food items, complete orders with minimal effort, and understand the instructions clearly. The system also responds quickly and works smoothly on both mobile and desktop devices. In general, the results show that the Darb e-Link System is efficient, reliable, and convenient for staff users, making it suitable for daily food ordering tasks.

Table 6
Darb e-Link System

5 – Strongly 4 – Agree 3 – Neutral 2 – Disagree 1 – Strongly Disagree

Faculty	5	4	3	2	1	TWP	WM	Descr.	Rank
1. The Darb e-Link System is easy to learn and understand.	9	7	0	0	0	16	4.56	SA	6
2. I can navigate the Darb e-Link System without any difficulty.	12	4	0	0	0	16	4.75	SA	3
3. The layout and functions of the system are simple and user-friendly.	12	4	0	0	0	16	4.75	SA	3
4. I can easily find the food items or options I need in the system.	13	2	1	0	0	16	4.75	SA	3
5. The ordering process in the Darb e-Link System is straightforward and clear.	13	3	0	0	0	16	4.81	SA	2
6. I can complete my food order with minimal steps or effort.	11	4	1	0	0	16	4.62	SA	5
7. The instructions and labels in the system are easy to understand.	9	6	1	0	0	16	4.50	SA	7
8. The system responds quickly when I click or submit actions.	11	5	0	0	0	16	4.69	SA	4
9. I can use the Darb e-Link System without needing help from others.	12	4	0	0	0	16	4.75	SA	5
10. The system works smoothly and consistently across both mobile and desktop	16	0	0	0	0	16	5.00	SA	1

devices.									
						Total	**4.718**	**SA**	

Overall, the faculty respondents strongly agreed with the usability of the Darb e-Link System, as shown by the total weighted mean of 4.718, interpreted as Strongly Agree (SA). This indicates that faculty members find the system very easy to learn and navigate, with a simple and user-friendly layout. They can easily find food items, complete orders with minimal effort, and understand the instructions clearly. The system also responds quickly and works very smoothly on both mobile and desktop devices. In general, the results show that the Darb e-Link System is highly efficient, reliable, and convenient for faculty users.

Conclusion

The Darb e-Link System successfully improved the old manual ordering process. It made ordering faster, reduced mistakes, and helped customers place orders easily using their phones. The admin can now manage menus, check orders, and update status more smoothly. Overall, the system made food ordering easier and more organized for both the customer and the seller.

The survey results indicate that the Darb e-Link System is highly acceptable and user-friendly, as reflected by the Strongly Agree (SA) ratings from all respondents. The overall weighted means for the total respondents (4.56), students (4.46), staff (4.564), and faculty (4.718) show that users find the system easy to learn, simple to navigate, and efficient to use.

Respondents agreed that food items are easy to find, the ordering process is clear and requires minimal effort, and the system can be used independently without assistance. In addition, the system responds quickly and functions smoothly across both mobile and desktop devices.

Overall, the Strongly Agree results confirm that the Darb e-Link System is

effective, reliable, and suitable for food ordering purposes, successfully improving the traditional manual ordering process and enhancing user experience for students, staff, and faculty.

Introductory Sentence

- Connect the system directly to GCash or Maya so payments are verified faster without manual checking.
- Create simple reports showing daily or monthly sales to help the admin track income and best-selling items.
- Add a push notification feature for the admin so they will immediately receive an alert when a new order comes in. This helps them respond faster and improves overall service speed.
- Mobile app.

REFERENCES

Acharya, K. (2024). Online Food Order System. Available at SSRN 4814732.

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4814732(Accessed Date: August 2025)

Alcalde, J. R., Malabanan, M. A., Hernandez, P. G., & Jamoralin, M. L. (2022). Customer satisfaction of DLSUD students on online food service delivery (Undergraduate thesis, De La Salle University – Dasmariñas). International Journal of Thesis Projects and Dissertations, 10(3), 26–40.
https://doi.org/10.5281/zenodo.6856224 (Accessed Date: August 2025)

Arias, C. A. R., & Bayna-Mariano, R. I. (2024). Online food delivery service satisfaction, food choices, and usage frequency among UPLB students aged 18-24 during the COVID-19 pandemic. Journal of Human Ecology and Sustainability, 2(1), 4.
https://ageconsearch.umn.edu/record/339944/ (Accessed Date: August 2025)

Chakraborty, D. (2019). Customer satisfaction towards food service apps in Indian metro cities. FIIB Business Review, 8(3), 245–255.
https://doi.org/10.1177/2319714519844651 (Accessed Date: August 2025)

Egereonu, S. K. (2024). Optimized Web-based Online Food Ordering System: Design and Implementation. International Journal of Computer Science Languages, 2(2), 1-28.
https://research-reels.com/publication/optimized-web-based-online-food-ordering-system-design-and-implementation/ (Accessed Date: August 2025)

Harbinger Group. (2023, March 16). The 4D development methodology for modernization. Harbinger Group.
https://www.harbingergroup.com/blogs/the-4d-development-methodology-for-modernization/ (Accessed Date: August 2025)

Jiangmin, L. (2024). Development of a Web-based Food Ordering System. Asia Pacific Journal of Management and Sustainable Development, 12(2 Part 3), 1-1.
https://www.ejournals.ph/article.php?id=25951 (Accessed Date: August 2025)

Peng, B., Erkoc, M., & Omachonu, V. K. (2024). Online Food Ordering and Delivery: A Study on the Use of Customer Service Data and Quality Function Deployment. Journal of Food Distribution Research, 55(2). https://www.fdrsinc.org/wp-content/uploads/2024/08/JFDR-July-2024-Full-Issue.pdf#page=69 (Accessed Date: August 2025)

Ray, A., Dhir, A., Bala, P. K., & Kaur, P. (2019). Why do people use food delivery apps (FDA)? A uses and gratification theory perspective. Journal of Retailing and Consumer Services, 51, 221–230. https://doi.org/10.1016/j.jretconser.2019.05.025 (Accessed Date: August 2025)

Sethu, H. S., & Saini, B. (2016, July 15–17). Customer perception and satisfaction on ordering food via Internet: A case on Foodzoned.com, in Manipal [Conference paper]. In Proceedings of the Seventh Asia-Pacific Conference on Global Business, Economics, Finance and Social Sciences, Kuala Lumpur, Malaysia. Retrieved from Scribd: https://www.scribd.com/document/429617703/Food-Zoned (Accessed Date: August 2025)

APPENDICES

APPENDIX A

TEST CASES

The Darb-eLink System went through different test cases to ensure that all software components were functioning properly. The functionality tests confirmed that the admin login system, online menu, cart management, order placement, payment processing, and SMS notification features were working correctly. The system accurately processed delivery and pickup orders, updated order statuses, and recorded all transactions in the database. SMS notifications were sent properly whenever the admin updated the order status. Data recording was also verified to ensure that all customer details, orders, and payment proofs were saved with complete and accurate information.

Further testing focused on reliability, usability, efficiency, compatibility, load, stress, and beta performance. The system remained stable during multiple simultaneous users and continuous operations, with no crashes or major errors observed. It handled network interruptions properly and resumed operations without losing data. Page loading speed and dashboard response time met expected standards. The system was also compatible with different devices, browsers, screen sizes, and operating systems. During peak load and stress testing, the Darb-eLink System continued to operate with only minor slowdowns. Overall, beta testing confirmed that the system is reliable, user-friendly, and capable of handling real-time ordering and administrative tasks efficiently.

1.Functionality Test Cases

Test Case ID	Test Description	Steps	Expected Result	Actual Output
FUN-01	Admin valid login	Enter correct username & password	Admin successfully logs in and is redirected to dashboard	Admin successfully logged in and was redirected to the dashboard without errors.
FUN-02	Admin invalid login	Enter incorrect admin credentials	Error message appears: “Invalid credentials”	System displayed “Invalid credentials” and prevented login.
FUN-03	Customer views food menu	Click the shared link → Menu loads	Menu items display with price, image, and categories	Menu loaded correctly with complete images, prices, and category list.
FUN-04	Add item to cart	Select item → Set quantity → Add to cart	Item successfully appears in cart	Item was added to the cart and displayed with the correct quantity.
FUN-05	Remove item / adjust quantity	Open cart → Increase/decrease quantity or remove item	Cart updates values correctly	Cart quantities updated correctly; removed items disappeared immediately.
FUN-06	Place order (Delivery)	Fill checkout form → Select Delivery → Upload payment proof → Confirm	Order submitted; system records customer details & payment receipt	Order was submitted successfully, and payment proof was uploaded and saved.
FUN-07	Place order (Pickup)	Fill checkout form → Select Pickup → Choose payment → Confirm	Order submitted and marked as pickup	Pickup order was submitted and marked as “For Pickup” in the admin panel.
FUN-08	10-second cancellation	Place order → Cancel within 10 seconds	Order is cancelled before final submission	Order was cancelled before submission and did not appear in the orders list.
FUN-09	Order auto-submit after timer	Do not cancel during timer, press cancel button if you want add more items.	Order is automatically placed	After the countdown, the order was automatically submitted to the system.
FUN-10	Admin add product	Navigate to Menu → Add item with	Product appears in product list	New product appeared in the item list with correct name, price,

		details → Save		and image.
FUN-11	Admin edits product	Select edit → Update name/price/image → Save	Product updates successfully	Product details were updated and reflected instantly in the menu.
FUN-12	Admin deletes product	Click delete → Confirm	Product removed from menu	Product was successfully deleted and no longer displayed in the menu.
FUN-13	Admin updates order status (Delivery)	Set status to Confirmed / Out for Delivery / Delivered	Status updates and SMS sent	Status updated properly and customer received the correct SMS notification.
FUN-14	Admin updates order status (Pickup)	Set status to Confirmed / Ready for Pickup / Picked Up	Status updates and SMS sent	Pickup order status updated correctly and SMS was sent to customer.
FUN-15	Customer receives SMS notification	Wait for admin update → Check phone	SMS received based on order status	Customer received the accurate SMS notification instantly.
FUN-16	Order history page	Customer → Order History	System displays customer's previous orders	Customer's past orders appeared correctly with order details.
FUN-17	Upload payment proof (GCash/Maya)	Select file → Submit	File uploaded and saved to server	Payment receipt was uploaded successfully and stored in the system.
FUN-18	COD proof upload	Delivery personnel uploads proof → Submit	Proof saved; order marked completed	COD proof image uploaded successfully and order marked as completed.
FUN-19	System handles multiple customers	10+ simultaneous menu access	System loads without errors	System handled multiple users without lag, timeout, or loading issues.

2. Reliability Testing

Test Case ID	Test Description	Steps	Expected Result	Actual Output
REL-01	Consistency check on ordering	Place 20 consecutive orders using the customer link	System processes all orders correctly	All 20 orders were successfully submitted and recorded in the admin dashboard

				without errors.
REL-02	Network interruption during checkout	Disconnect Wi-Fi or mobile data while placing an order	System pauses and resumes normally	System displayed a "Network Error" message and continued the process after reconnection without losing entered customer information.
REL-03	Admin session reliability	Leave admin dashboard idle for several minutes	Admin remains logged in or safely logged out	System maintained the session for a period of time and did not lose any unsaved data.
REL-04	SMS status reliability	Update order status repeatedly (Confirmed → Delivered or Confirmed → Picked Up.)	System sends correct SMS once per update	System sent accurate SMS notifications aligned with each status change, no duplicate SMS recorded.

3 . Usability Testing

Test Case ID	Test Description	Steps	Expected Result	Actual Output
USA-01	Navigation clarity	New user opens the link and navigates the system	User completes tasks	New user was able to browse the menu, add items to cart, and place an order without confusion.
USA-02	Button visibility	Check menus, buttons, and clickable elements	All clickable	All buttons such as "Add to Cart," "Checkout," "Submit Order," and "Edit/Delete" (admin) were clearly visible and responsive.
USA-03	Error message clarity	Trigger input errors (empty fields, invalid contact number, no receipt uploaded)	Readable message	System displayed clear and readable error messages such as "Please enter your name," "Invalid contact number," and "Payment proof required."

USA-04	Order placement ease	New user places an order (pickup or delivery)	Completed within 1–2 minutes	User completed the entire ordering process (menu → cart → checkout → submit) within 1 minute and 20 seconds.

4. Efficiency Testing

Test Case ID	Test Description	Steps	Expected Result	Actual Output
EFF-01	Page loading speed (Customer Menu)	Open the customer link → Menu loads	Loads within 1–3 seconds	Menu page loaded in 1.8 seconds on a stable connection; slight delay observed on slower internet.
EFF-02	Dashboard load speed	Admin logs in Dashboard loads	Loads 3 seconds	Dashboard loaded in 2.1 seconds on normal connection; minor delay noted during peak traffic or slow network.
EFF-03	High user load performance	Simulate 10 or more simultaneous users accessing the menu	System remains stable	System handled multiple users smoothly with no crashes; slight loading delay occurred under slow network conditions.

5. Compatibility Testing

Test Case ID	Test Description	Steps	Expected Result	Actual Output
POR-01	Cross-Browser Compatibility	The system was opened using Google Chrome, Mozilla Firefox, and Microsoft Edge.	The system should behave the same in all browsers.	The Darb e-Link System worked properly on all tested browsers with no layout or function issues.
POR-02	Device Compatibility	The system was accessed using a laptop and a desktop computer and mobile devices.	The system should work on both devices.	The system functioned correctly on both laptop and desktop without errors.
POR-03	Screen Resolution Testing	The screen size was changed to different	The interface should adjust	The system layout adjusted properly

		resolutions.	automatically.	and remained clear and usable on different screen sizes.
POR-04	Operating System Compatibility	The system was tested on Windows and web browsers.	The system should run consistently on both operating systems.	The system operated smoothly on both Windows with the same features and performance.

6. Load Testing Procedure

Scenario ID	Activity	Number of Users	Objective	Expected Outcome	Actual Output
LT-01	Admin Login Simultaneously	admin	Check login performance during busy time	All admin can log in with fast response time.	Admin logged in successfully with response time below 3 seconds.
LT-02	Multiple Order Placement	50–100 customers	Test system speed while placing many orders	Orders are saved correctly and processed fast.	Orders were placed successfully and recorded accurately in the system.
LT-03	Order Management and Monitoring	admin	Test admin dashboard during heavy use	Orders load correctly with no errors.	Admin dashboard remained responsive and displayed orders properly.
LT-04	Peak Load Scenario	100–150 customers	Test system performance during peak hours	System handles many users with minor slowdowns allowed.	The system remained operational with slight slowdowns but no crashes.

Scenario ID	Activity	Number of Users	Objective	Expected Outcome	Actual Output
LT-05	Mixed Workload	100 customers ordering + admin managing orders	Test combined customer and admin activities	System stays stable while all actions are running.	All customer orders and admin actions were completed successfully while the system stayed stable.

7. Stress Testing Procedure

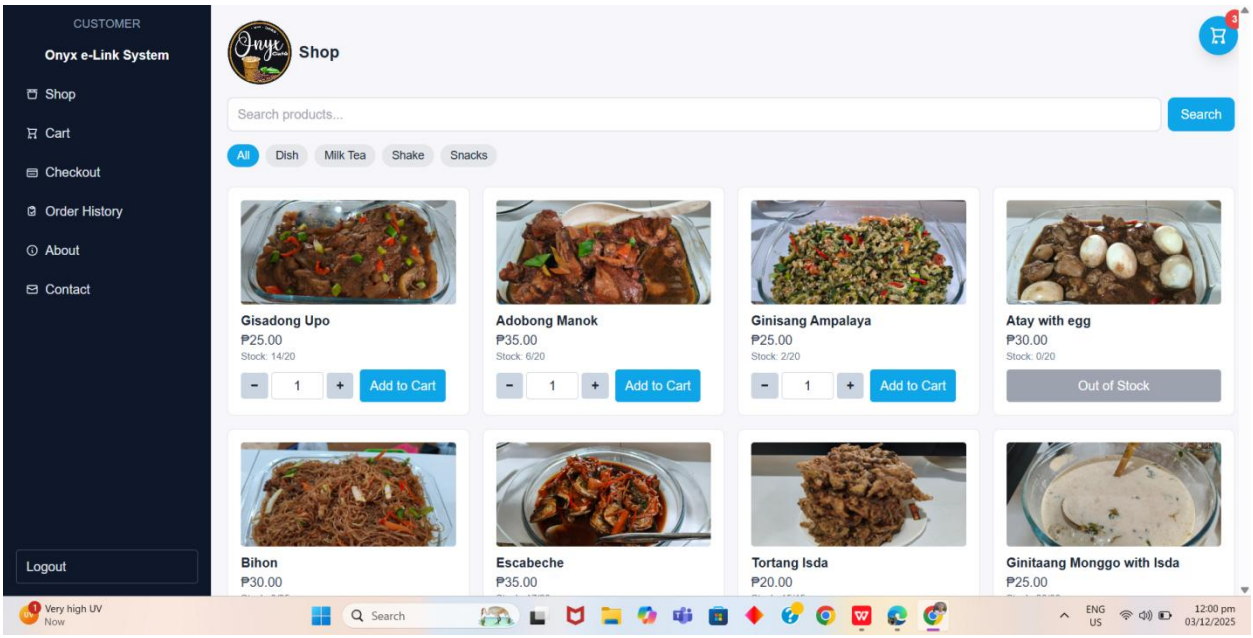
Scenario ID.	Activity	Objective	Expected Outcome	Actual Output (Darb e-Link System)
ST-01	Simulate 150+ simultaneous users logging in/opening the Darb link	Determine the system's concurrent access capacity	The system should load normally with minimal delay and no crash	The system remained accessible and stable. Page loading was slightly slower at peak user count but no crash occurred; login and dashboard functioned correctly.
ST-02	150 users performing Time-In/Time-Out (simulated ordering actions) at the same time	Check responsiveness when many users submit actions simultaneously	*S*ystem should continue saving logs and processing actions with acceptable response time	Orders were still recorded correctly. Some delays of 1–3 seconds appeared when adding items to cart, but all transactions were saved and processed successfully.
ST-03	Continuous scanning/ordering for **2 hours**	Test stability under continuous use	No memory leaks; system stays responsive during long usage	The system remained stable for 2 hours. No freezing or memory errors occurred. Only minor slowdowns appeared when many images loaded repeatedly, but the site remained functional.

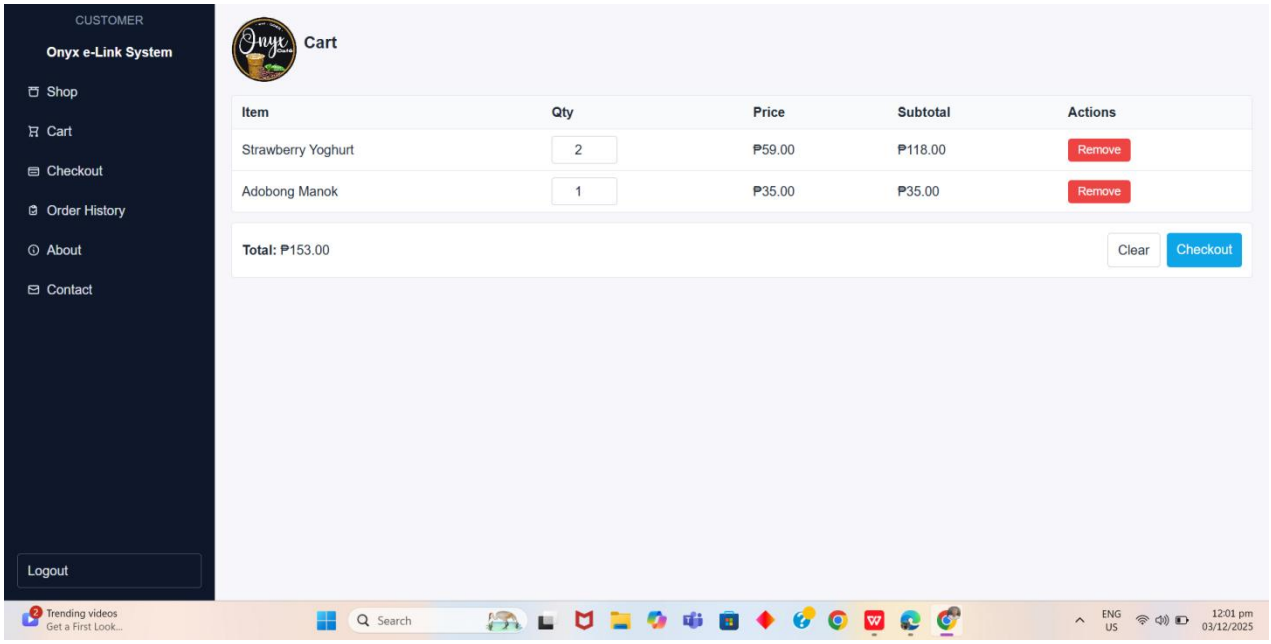
8. Beta Testing

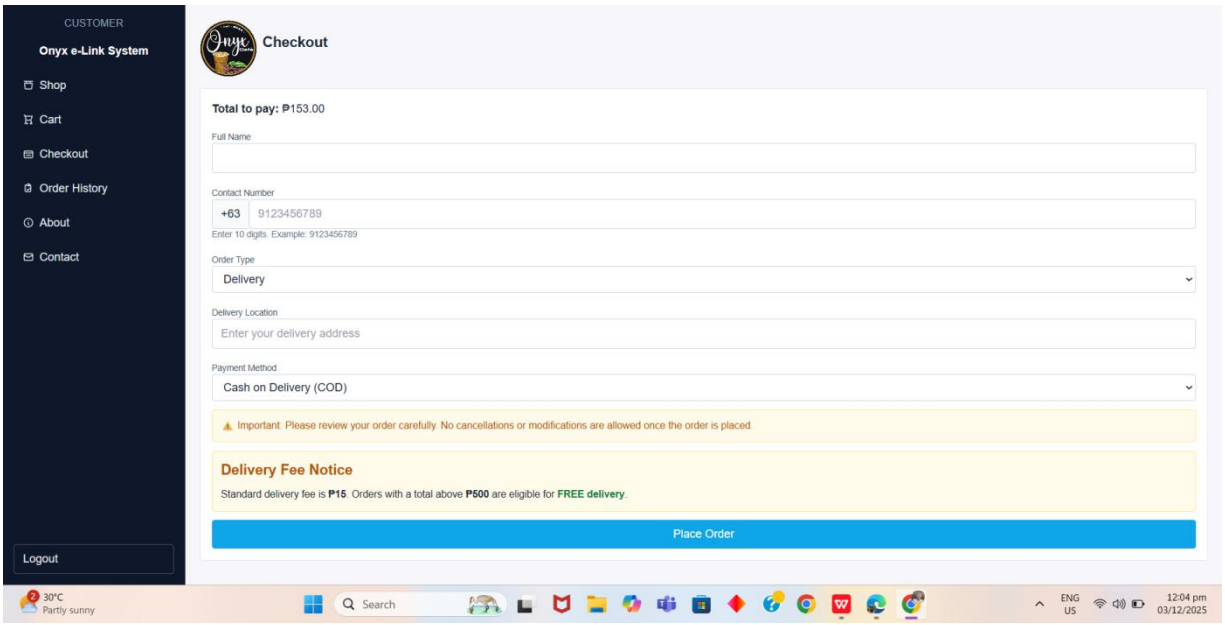
Test ID.	Main FeatureTested	Activity Performed	Expected Result	Actual Result
BT-01	Online Menu Access	Open shared link to view menu	Menu loads correctly	Menu displayed correctly
BT-02	Add to Cart	Select items and add to cart	Items added with correct quantity	Items added successfully
BT-03	Order Placement	Place order for pickup or delivery	Order submitted successfully	Orders submitted without errors
BT-04	Payment Option	Select GCash, Maya, or COD	Payment option accepted	Payment options worked properly
BT-05	Receipt Upload	Upload payment receipt	Receipt uploaded successfully	Receipt saved correctly
BT-06	Order Status Update	Admin updates order status	Status updated in system	Status updated correctly
BT-07	SMS Notification	System sends SMS updates	Customer receives SMS	SMS received successfully
BT-08	Menu Management	Admin adds/edits/deletes menu	Menu updated instantly	Changes reflected correctly
BT-09	Order Monitoring	Admin views customer orders	Orders displayed clearly	Orders shown accurately

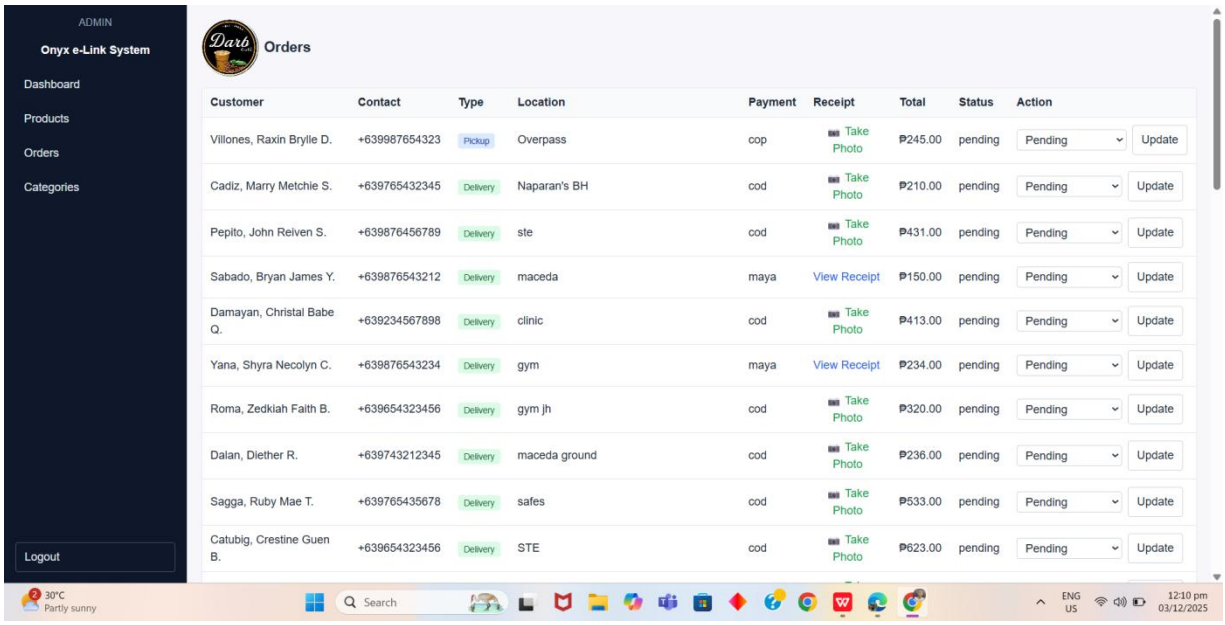
APPENDIX B

SAMPLE INPUT AND OUTPUT REPORT






CUSTOMER
Onyx e-Link System
Shop
Cart
Checkout
Order History
About
Contact
Logout
Checkout
Total to pay: ₱153.00
Full Name
Contact Number
+63
9123456789
Enter 10 digits. Example: 9123456789
Order Type
Delivery
Delivery Location
Enter your delivery address
Payment Method
Cash on Delivery (COD)
Important: Please review your order carefully. No cancellations or modifications are allowed once the order is placed.
Delivery Fee Notice
Standard delivery fee is ₱15. Orders with a total above ₱500 are eligible for FREE delivery.
Place Order


ADMIN
Onyx e-Link System
Dashboard
Products
Orders
Categories
Logout
Orders
Customer
Contact
Type
Location
Payment
Receipt
Total
Status
Action
Villones, Raxin Brylle D.
+639987654323
Pickup
Overpass
cop
Take Photo
₱245.00
pending
Pending
Update
Cadiz, Marry Metchie S.
+639765432345
Delivery
Naparan's BH
cod
Take Photo
₱210.00
pending
Pending
Update
Pepito, John Reiven S.
+639876456789
Delivery
ste
cod
Take Photo
₱431.00
pending
Pending
Update
Sabado, Bryan James Y.
+639876543212
Delivery
maceda
maya
View Receipt
₱150.00
pending
Pending
Update
Damayan, Christal Babe Q.
+639234567898
Delivery
clinic
cod
Take Photo
₱413.00
pending
Pending
Update
Yana, Shyra Necolyn C.
+639876543234
Delivery
gym
maya
View Receipt
₱234.00
pending
Pending
Update
Roma, Zedkiah Faith B.
+639654323456
Delivery
gym jh
cod
Take Photo
₱320.00
pending
Pending
Update
Dalan, Diether R.
+639743212345
Delivery
maceda ground
cod
Take Photo
₱236.00
pending
Pending
Update
Sagga, Ruby Mae T.
+639765435678
Delivery
safes
cod
Take Photo
₱533.00
pending
Pending
Update
Catubig, Crestine Guen B.
+639654323456
Delivery
STE
cod
Take Photo
₱623.00
pending
Pending
Update

APPENDIX C

Documentation



APPENDIX D

Survey Questionnaire

Perceived Ease of Use (PEOU)

Directions:

Please indicate your level of agreement with each statement using the scale below:

5 – Strongly 4 – Agree 3 – Neutral 2 – Disagree 1 – Strongly Disagree

Agree

Part I. Respondent Profile

- Role: [ ] Student [ ] Staff [ ] Faculty [ ] Vendor
- Age: _____
- Sex: [ ] Male [ ] Female [ ] Prefer not to say
- Frequency of using the Darb e-Link System: [ ] Daily [ ] Weekly [ ] Monthly [ ] Rarely

No.	Statement	5 - Strongly	4 – Agree	3 - Neutral	2 - Disagree	1 – Strongly Disagree
1	The Darb e-Link System is easy to learn and understand.					
2	I can navigate the Darb e-Link System without any difficulty.					
3	The layout and functions of the system are simple and user-friendly.					
4	I can easily find the food items or options I need in the system.					
5	The ordering process in the Darb e-Link System is straightforward and clear.					
6	I can complete my					

	food order with minimal steps or effort.					
7	The instructions and labels in the system are easy to understand.					
8	The system responds quickly when I click or submit actions.					
9	I can use the Darb e-Link System without needing help from others.					
10	The system works smoothly and consistently across both mobile and desktop devices.					

APPENDIX E



CURRICULUM VITAE

Personal Data

Name	: Ian Christoptopher Tano Roda
Gender	: Male
Date of Birth	: April 02, 2003
Place of Birth	: Margosatubig Zamboanga del Sur
Civil Status	: Single
Citizenship	: Filipino
Religion	: Seventh-Day Adventist Church
Address	: Sitio Buli, Margosatubig Zamboanga del Sur
Father's Name	: Marcial Iligan Roda
Occupation	: OFW
Present Address	: Poblacion Guipos, Zamboanga del Sur
Mother's Name	: Rosalie T. Roda
Occupation	: OFW

EDUCATIONAL BACKGROUND

Education Graduated	**School**	**Year**
Primary	: Margosatubig Regional Pilot School	2014-2015
Junior High School	: Toribio Minor National High School	2017-2018
Senior High School	: Kalian National High School	2021-2022
Tertiary	: JHCSC Main Campus	2025-2026
Degree	: Bachelor of Science in Information Technology (BSIT)	

CURRICULUM VITAE



Personal Data

Name	: Junelie Fuentes Labad
Gender	: Male
Date of Birth	: November 06, 2003
Place of Birth	: Dimataling, Zamboanga Del Sur
Civil Status	: Single
Citizenship	: Filipino
Religion	: The Kingdom Of God and His Righteousness
Address	: Poblacion, San Miguel Zamboanga Del Sur
Father's Name	: Maximino Viña Labad jr
Occupation	: Passenger Tricycle Driver
Present Address	: Poblacion San Miguel, Zamboanga del Sur
Mother's Name	: Sally Fuentes Labad
Occupation	: House Wife

EDUCATIONAL BACKGROUND

Education Graduated	**School**	**Year**
Primary	: San Miguel Central Elementary School	2014-2015
Junior High School	: Betinan National High School	2017-2018
Senior High School	: Betinan National High School	2021-2022
Tertiary	: JHCSC Main Campus	2025-2026
Degree	: Bachelor of Science in Information Technology (BSIT)	

CURRICULUM VITAE



Personal Data

Name	: Melvin Adlaon Porferio
Gender	: Male
Date of Birth	: November 17, 2003
Place of Birth	: Baluno,Dimataling Zamboanga del Sur
Civil Status	: Single
Citizenship	: Filipino
Religion	: Iglisia Filipina Independente (IFI)
Address	: Tiparak Tambulig ,Zamboanga del Sur
Father's Name	: Ruben Quirog Porferio
Occupation	: Security Guard
Present Address	: Mati SanMiguel,, Zamboanga del Sur
Mother's Name	: Emely Adlaon Porferio
Occupation	: HouseWife

EDUCATIONAL BACKGROUND

Education Graduated	**School**	**Year**
Primary	: Baluno Elementary School	2014-2015
Junior High School	: Molave Vocational Technical School	2017-2018
Senior High School	: Molave Vocational Technical School	2021-2022
Tertiary	: JHCSC Main Campus	2025-2026
Degree	: Bachelor of Science in Information Technology (BSIT)	

CURRICULUM VITAE



Personal Data

Name	: Mark Cloyd Alcala Cabog
Gender	: Male
Date of Birth	: August 02, 2003
Place of Birth	: Liloy Zamboanga del Norte
Civil Status	: Single
Citizenship	: Filipino
Religion	: CAMACOP
Address	: Old Mirapao Dinas, Margosatubig Zamboanga del Sur
Father's Name	: Basilio Lasta Cabog
Occupation	: Minister
Present Address	: Old Mirapao Dinas, Margosatubig Zamboanga del Sur
Mother's Name	: Rebacca Villafuerte Alcala
Occupation	: Lady Minister

EDUCATIONAL BACKGROUND

Education Graduated	**School**	**Year**
Primary	: Dansal Elementary School	2014-2015
Junior High School	: Lapuyan National High School	2017-2018
Senior High School	: Dinas National High School	2021-2022
Tertiary	: JHCSC Main Campus	2025-2026
Degree	: Bachelor of Science in Information Technology (BSIT)	